

1 Towards a FAIR Framework for Earthquake catalogues in
2 Aotearoa New Zealand

3 A Community White Paper

4 *Making New Zealand's earthquake catalogues Findable, Accessible, Interoperable, and Reusable*

5 Kenny Graham¹, Jonathan Hanson¹, Calum Chamberlain²,
Eleanor Mestel², Chris Rollins¹, Sandra Bourguignon¹,
Florent Aden¹, Annemarie Christophersen¹,
Codee-Leigh Williams², and Brendon Bradley³

¹ Earth Sciences New Zealand

² Victoria University of Wellington

³ University of Canterbury

6 July 2026

Executive Summary

Earthquake catalogues underpin seismology, earthquake forecasting, and the seismic hazard models used in engineering and risk management. Aotearoa New Zealand has produced many earthquake catalogues through decades of operational monitoring, scientific research, hazard assessment, and temporary seismic deployments. Collectively, these catalogues represent a nationally significant information asset built through sustained public investment, scientific effort, and observational infrastructure. However, as a national collection, they fall short of the FAIR principles of being *Findable*, *Accessible*, *Interoperable*, and *Reusable*: there is no common index, so they are hard to *find*; few carry the persistent identifiers and versioning needed to *access* and cite a specific release; they are built to different standards and conventions, so they are not readily *interoperable*; and inconsistent documentation limits *reuse*. Earthquake catalogues also evolve naturally over time as seismic networks expand, instrumentation changes, processing methods improve, and scientific understanding advances, increasing the importance of transparent metadata, provenance, and version control.

The cost of the current situation is concrete: the 2022 National Seismic Hazard Model had to re-standardise magnitudes across sources before it could use them (Gerstenberger et al., 2024; Christophersen et al., 2024), and assembling an integrated national earthquake catalogue for Aotearoa New Zealand likewise required a dedicated effort to reconcile and harmonise multiple source catalogues (Rollins et al., 2025); combining catalogues remains a manual, error-prone task. A companion inventory (1960–2026) documents 93 of these catalogues, across national, regional, temporary, historical, and derived categories.

Two national workshops (2023 and 2024) brought together stakeholders from universities, agencies, and industry to identify needs and shape a practical framework guided by both the FAIR principles and the CARE principles (Collective benefit, Authority to control, Responsibility, and Ethics). This white paper sets out the current landscape, the community-identified gaps, and a proposed national framework: a “catalogue of catalogues” that improves findability and supports a minimum standard for easier interoperability. The framework makes catalogues findable through a common submission profile and provides FAIR services (including quality assessment, merge, analytics, versioning, and programmatic access) for the catalogues it coordinates. The framework is accompanied by proposed governance arrangements, a roadmap, and an indicative resourcing envelope. A working prototype demonstrates the core workflows, from loading and merging catalogues to analysis and export in several formats. The framework proposed here offers practical pathway towards a more discoverable, interoperable, and sustainable earthquake-catalogue ecosystem and is released for community review and refinement toward a sustained national capability.

Key Recommendations

1. Adopt a minimum-attributes *submission profile* and a controlled parameter lexicon, aligned with the international QuakeML and FDSN seismic-data standards, so every catalogue is described consistently (distinguishing required from recommended fields).
2. Establish the openly versioned “catalogue of catalogues” as the low-barrier, near-term community deliverable, beginning with discovery: a searchable index of submission profiles that records which catalogues exist and where to find them.
3. Build out the FAIR services incrementally (a programmatic interface, cross-agency event-identifier mapping, transparent merge, configurable quality scoring, and analytics) for the catalogues it coordinates.
4. Require persistent identifiers (DOIs), explicit versioning, and machine-readable changelogs for DOI-bearing catalogue releases, so any cited catalogue version is reproducible.
5. Represent uncertainty explicitly: report uncertainty values where available, and where they are absent record that absence or a qualitative confidence class, so that no record implies more precision than the data support.

57 6. Establish formal governance (a Steering Committee with substantive Māori representation,
58 and a Data Stewardship Team) that operationalises both FAIR and CARE principles,
59 underpinned by sustained multi-year funding.

60 **Keywords:** earthquake catalogues; FAIR principles; CARE principles; seismology; research data
61 infrastructure; data governance; Aotearoa New Zealand.

62 Contents

63	1 Introduction: The Problem and the Opportunity	6
64	2 The Current Landscape and International Benchmark	8
65	3 A Community-Driven Process	19
66	3.1 Prototype Portal: Proof of Concept	19
67	4 The Proposed FAIR Framework	20
68	4.1 Metadata Standards and a Shared Lexicon	22
69	4.2 Catalogue Submission Profile (Minimum Attributes)	22
70	4.3 Interoperability: Programmatic Access and Cross-Catalogue Identifiers	24
71	4.4 Catalogue Merge	24
72	4.5 Data Quality Assessment	26
73	4.6 Versioning, Persistent Identifiers, and Provenance	27
74	4.7 Special Catalogue Types	28
75	4.8 Licensing and Access; FAIR Compliance Indicators	28
76	4.9 Aotearoa New Zealand Data Governance: FAIR with CARE	28
77	5 Implementation: Governance, Roadmap, and Funding	29
78	5.1 Governance, Roles, and Responsibilities	29
79	5.1.1 Steering Committee	29
80	5.1.2 Data Stewardship Team	29
81	5.1.3 Catalogue Contribution Process	30
82	5.1.4 Catalogue Custody, Updates, and Use Conditions	30
83	5.1.5 Success Metrics and Annual Reporting	30
84	5.2 Roadmap	31
85	5.3 Funding and Resourcing	31
86	6 Benefits by Stakeholder	32
87	7 Risks and Limitations	32
88	8 Conclusion and Call to Action	33
89	Data and Code Availability	34

90	A International Benchmark Detail	34
91	A.1 New Zealand (GeoNet)	34
92	A.2 United States (USGS ComCat)	35
93	A.3 Switzerland (SED)	35
94	A.4 European and Global Aggregators	35
95	B Technical Specification	35
96	B.1 Metadata Field Hierarchy	35
97	B.2 Controlled Parameter Lexicon	38
98	B.3 QuakeML/FDSN Crosswalk	38
99	B.4 Versioning Scheme	40
100	B.5 API Design	40
101	B.6 Access Control and Audit	40
102	B.7 Uncertainty Representation	41
103	B.8 Quality Checks and Full Grade Scheme	41
104	B.9 Special Catalogue Type Handling	42
105	B.10 FAIR Compliance Matrix	43
106	C Summary of Workshop Themes	43
107	D Glossary	44

1 Introduction: The Problem and the Opportunity

Earthquake catalogues are the cornerstone of seismology and the shared evidence base for hazard models, engineering design, rapid response, and fundamental research (Woessner et al., 2010). Their value depends on being trustworthy, comparable, and reusable.

In Aotearoa New Zealand there are many different earthquake catalogues, reflecting decades of investment in operational monitoring, scientific research, hazard assessment, and temporary seismic deployments. The authoritative national earthquake catalogue forms part of the Earthquake Information Dataset (EID), one of New Zealand’s Nationally Significant Datasets and Collections. GeoNet operates the national seismic monitoring network and produces the national earthquake catalogue, providing broad coverage and open access through web and programmatic services (GeoNet, 2025a,c,d); although earthquake location quality and catalogue completeness vary across the country as a function of seismic station distribution (Warren-Smith et al., 2025). Also, metadata conventions have evolved over time.

Earthquake catalogues are also not static products. They naturally evolve over time as seismic networks expand, instrumentation is upgraded, processing techniques improve, and new scientific understanding becomes available. As a result, changes in earthquake locations, magnitudes, uncertainties, event classifications, and levels of completeness through time are expected and reflect improvements in observational capability rather than deficiencies in the catalogue. The challenge is therefore not to eliminate these differences, but to document them transparently so that catalogue evolution can be understood and catalogues remain interpretable, comparable, and reusable.

Alongside the national catalogue sit many catalogues that are not integrated with, and are often not discoverable through, the national system:

- regional and volcano-specific catalogues built with dense local networks;
- catalogues from dense temporary deployments, including offshore, with low magnitudes of completeness
- historical and pre-instrumental compilations from written and oral records, including Eiby’s annotated list of New Zealand earthquakes from 1460 (Eiby, 1968), the *Atlas of Iseismic Maps of New Zealand Earthquakes* (Downes, 1995), and revised magnitudes for 1901–1993 (Dowrick and Rhoades, 1998);
- overseas sources that aggregate or reprocess New Zealand events, such as the ISC Bulletin and ISC-GEM catalogue, the Global CMT catalogue, and USGS ComCat (International Seismological Centre, 2025a; Ekström et al., 2012; USGS, 2025); and
- derived catalogues built for hazard and risk modelling, such as the magnitude-homogenised catalogue prepared for the 2022 National Seismic Hazard Model (Christophersen et al., 2024).

These are not isolated exceptions. A companion inventory covering 1960–2026 identifies 93 such catalogues, spanning regional and volcano-seismic networks, temporary onshore and offshore deployments, aftershock responses, historical compilations, and hazard-model datasets; this inventory is not exhaustive. Volcano-seismic, slow-slip and tremor, and temporary onshore and offshore deployment catalogues together account for over half of the inventory (54 of 93 entries), yet temporary deployments in particular remain among the least discoverable through the national system (see Data and Code Availability).

Many of these catalogues are hard to locate: those published before journals required information on data availability, or held on personal or ad-hoc platforms (for example GitHub or Zenodo),

can become effectively lost once their authors move on or retire, a problem compounded by an ever-growing number of catalogues. Harmonising magnitudes is a further, distinct cost: because these catalogues use different formats, identifiers, and magnitude scales, combining them is manual and error-prone, and the 2022 National Seismic Hazard Model required a dedicated effort to re-standardise magnitudes across sources before they could be used (Gerstenberger et al., 2024; Christophersen et al., 2024). The consequences of the status quo can include lost researcher time, reduced reproducibility, and less transparent uncertainty in decisions that can affect public safety and major infrastructure.

These challenges are likely to increase in importance over time. The number and diversity of earthquake catalogues continue to grow, driven by temporary deployments, advances in seismic processing and machine-learning methods, and the increasing use of earthquake data in hazard and risk assessment. At the same time, valuable datasets may become difficult to locate, access, or interpret as projects conclude and researchers move institutions. Without a coordinated approach, the benefits of these investments risk becoming increasingly fragmented.

A thin coordinating layer (a “catalogue of catalogues”) can make these datasets findable, comparable, and citable. It combines a common submission profile that makes catalogues findable with a set of richer FAIR services (comparison, merging, and citable versions) for the catalogues it coordinates. Internationally, initiatives such as the European Plate Observing System (EPOS) Seismology service show the value of standardised data services and coordinated infrastructure (EPOS, 2025b,a).

This white paper synthesises outcomes from two national workshops and intersession working-group activity to propose such a framework, tailored to New Zealand. The proposed framework combines FAIR principles (making earthquake catalogues Findable, Accessible, Interoperable and Reusable) with CARE principles (ensuring data governance delivers Collective benefit, recognises Authority to control, demonstrates Responsibility and upholds Ethics), creating a system that is both technically robust and socially responsible.

There are three aims: (i) to characterise the landscape of New Zealand’s earthquake catalogues and benchmark it against mature international catalogues; (ii) to propose a common set of minimum attributes for describing and creating catalogues; and (iii) to set out a gap analysis and near-term community steps toward a coordinated, harmonised national catalogue capability for earthquake science and engineering.

The recommendations and framework presented here are intended for earthquake scientists, data custodians, hazard practitioners, research funders, and infrastructure providers responsible for the long-term stewardship of New Zealand’s earthquake information assets.

Collectively, New Zealand’s earthquake catalogues represent decades of public investment, scientific effort, and observational infrastructure. They underpin earthquake science, hazard assessment, engineering design, and risk-informed decision making. Ensuring their discoverability, interoperability, and long-term stewardship is therefore a matter of national scientific capability.

To address these challenges, the proposed framework has been shaped by practical needs identified through the 2023 and 2024 community workshops. Participants consistently highlighted the need for a clear overview of available catalogues, improved discoverability, the ability to compare and merge catalogues across sources and greater transparency regarding catalogues quality and content, and suitability for specific analyses (for example, phase picks versus earthquake hypocentres). Figure 1 contrasts the current with the proposed coordinated approach.

It is helpful to state what success looks like. Within five years of implementing the framework, we envisage that:

- researchers can quickly identify the catalogues relevant to a given study;

- catalogues from temporary deployments stay discoverable and usable long after their projects end;
- each catalogue version is citable through a DOI, and its results reproducible;
- differences between overlapping catalogues are documented and traceable to their sources;
- data custodians retain ownership and stewardship of their datasets; and
- hazard and risk projects spend less effort locating and harmonising data, and more on analysis.

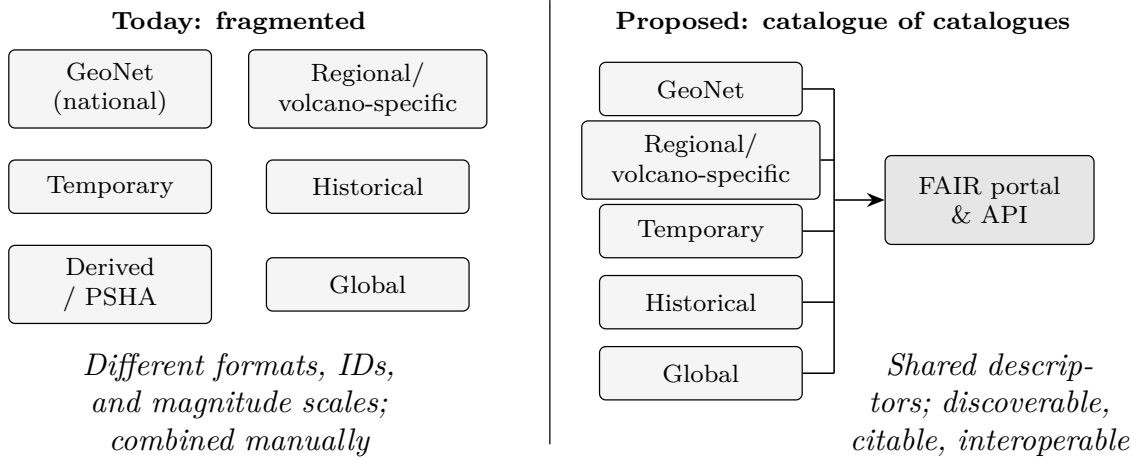


Figure 1: The problem and the opportunity (representative types shown; the full inventory lists 93 catalogues). Today, New Zealand’s earthquake catalogues are developed independently and combined manually. The proposed framework adds a coordinating layer that makes them interoperable through shared standards, a common portal, and a programmatic interface.

2 The Current Landscape and International Benchmark

New Zealand’s earthquake data ecosystem comprises several catalogue types that serve different purposes. The **national operational catalogue**, produced through GeoNet by Earth Sciences New Zealand (ESNZ; formerly GNS Science) and stewarded as part of the EID, provides the principal record of New Zealand seismicity and supports earthquake response, hazard assessment, engineering, and scientific research. Like earthquake catalogues worldwide, it has evolved over time as seismic networks, instrumentation, processing methods, and metadata practices have developed. Around it sit regional and volcano-specific catalogues offering high-resolution detail (Ilsley-Kemp and Mestel, 2025); often poorly discoverable temporary and aftershock deployments; specialised source-mechanism, slow-slip, palaeoseismic, and geodetic catalogues; relocated or otherwise method-specific research catalogues; and historical, derived, synthetic, and internationally aggregated compilations that are difficult to reuse without formal versioning, identifiers, and processing documentation (Gerstenberger et al., 2024). These groupings correspond to the working classification in Table 1. The ecosystem’s strengths lie in local precision and national consistency, while its gaps lie in completeness through time and space, processing consistency, magnitude and depth scales, metadata, transparency, and access to legacy and temporary data.

This ecosystem is large and heterogeneous. Table 1 sets out the working classification of catalogue types used in this paper. It draws on the community workshops but is our synthesis rather

225 than a transcript of them, and lists representative examples with characteristic strengths and
226 limitations.

Table 1: A working classification of earthquake-catalogue types in Aotearoa New Zealand used in this paper, informed by the community workshops, with representative (non-exhaustive) examples and references and characteristic strengths and limitations. The classification is indicative rather than definitive; categories are not mutually exclusive, and a given catalogue may fall under more than one type.

Type	Examples	Strengths / limitations
Regional / volcano-specific	Taupō, Whakaari, Ruapehu, geothermal fields	Precise depths, consistent processing within each network; limited spatial scope, weak interoperability
National operational	GeoNet rapid and reviewed, GeoNet centroid moment tensors (CMT), strong-motion datasets (GeoNet, 2025a)	Consistency, continuity, accessibility, manual review; evolving conventions, mixed magnitude types
Temporary deployments (onshore & offshore)	Alpine Fault OBS, Hikurangi (e.g. HOBITSS), Kaikōura/Cook Strait aftershocks, TVZ experiments (SAMBA, COSA, PTANGO) (Michailos et al., 2019; Yarce et al., 2019; Aziz Zanjani et al., 2021)	High-quality velocity models, precise locations and depths; short duration, limited discoverability
Source-mechanism	Ristau moment tensors, Global Centroid Moment Tensor (GCMT), finite-fault and stress-drop inversions (Ekström et al., 2012)	Kinematic and stress information for hazard and tectonics; standalone, weakly integrated
Historical / pre-instrumental	Pre-1940 compilations, oral and written records, NSHM inputs (Eiby, 1968; Downes, 1995; Gerstenberger et al., 2024)	Extend the record by centuries; large uncertainties, poor depth/location, few small events
Slow-slip & tremor	Slow-slip event detections, tectonic tremor and low-frequency earthquakes (LFEs) (Wallace and Beavan, 2010)	Capture aseismic slip and deep fault processes not seen in standard catalogues; individual events are transient (days–weeks), with imprecise timing
Palaeoseismic & geodetic	Palaeoseismic records, offshore (seafloor) geodetic detections (Fasola et al., 2023)	Extend the deformation and rupture history over centuries to millennia; sparse coverage, large timing uncertainties
Research / method-specific	Template-matching, machine-learning, relocation, and waveform-inversion catalogues (Michailos et al., 2025; Chamberlain et al., 2025; Eberhart-Phillips and Reyners, 2023)	Gains in completeness and precision, documented methods; bespoke formats, variable access
Derived / international	Point-in-time catalogue snapshots prepared for the NSHM; ISC, ComCat, EMSC combined catalogues (USGS, 2025; International Seismological Centre, 2025a; European-Mediterranean Seismological Centre, 2025b)	Broad or standardised coverage; not NZ-specific, variable completeness

The inventory is also spatially uneven (Figure 2). Panel (a) shows the footprint of each regional catalogue (a representative bounding box) over the New Zealand coastline and active-fault network, coloured by type; the 18 national or NZ-wide catalogues are not drawn because they span the whole country, and two Kermadec-arc catalogues lie north of the map. Panel (b) counts how many regional catalogues overlap each 0.1° cell, so the intensively studied areas stand out, the Taupō Volcanic Zone, the northern Hikurangi margin, and the central Alpine Fault, while large offshore tracts, Fiordland, Canterbury, and the far south remain thinly catalogued. This pattern of densely catalogued zones beside large under-served tracts is the spatial counterpart of the fragmentation a coordinating framework is designed to close.

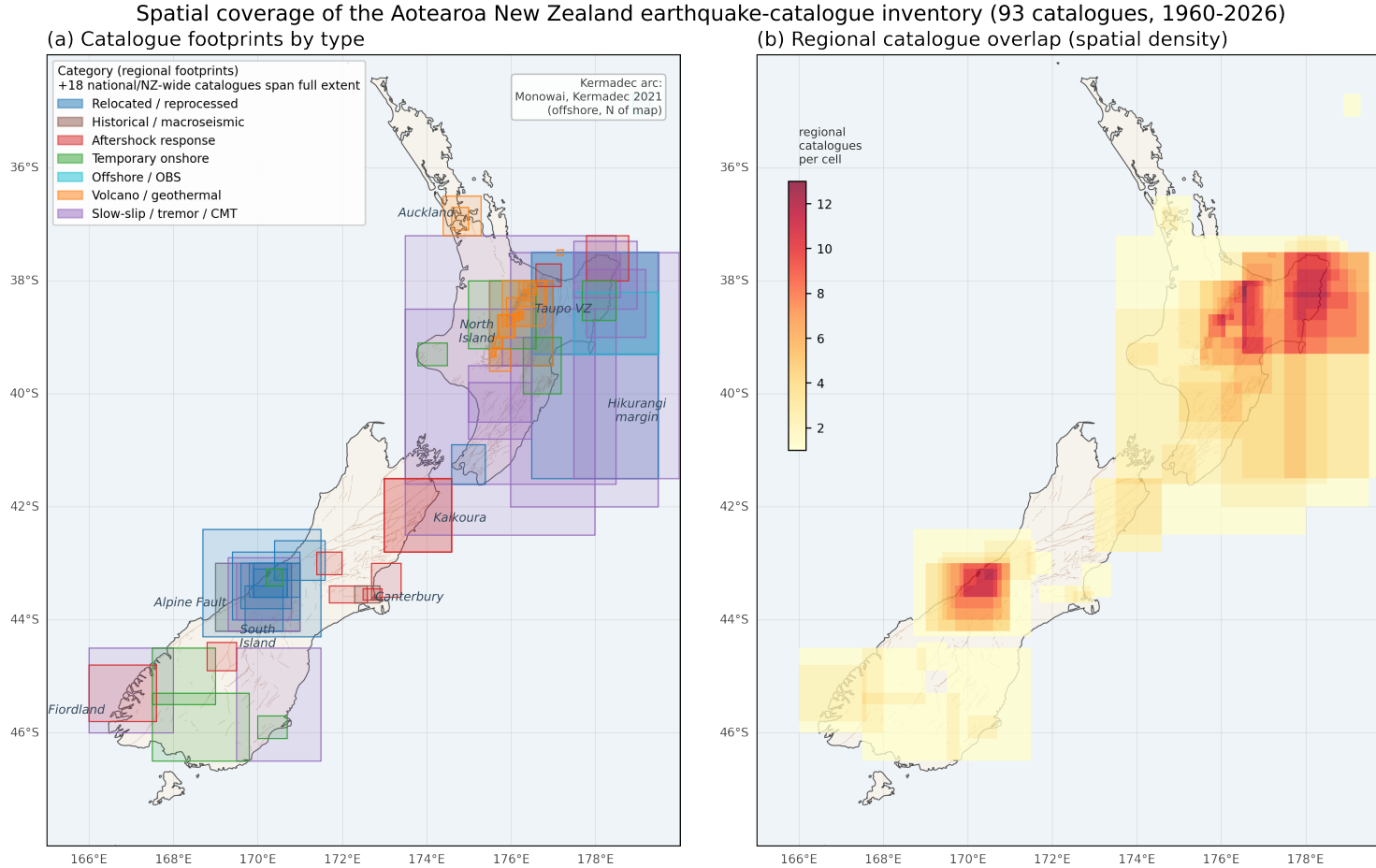


Figure 2: Spatial coverage of the New Zealand earthquake-catalogue inventory (93 catalogues, 1960–2026). (a) Footprint of each regional catalogue (a representative bounding box) over the coastline and active faults, coloured by type; 18 national or NZ-wide catalogues span the full extent and are not drawn, and two Kermadec-arc catalogues lie north of the map. (b) Number of regional catalogues overlapping each 0.1° cell, highlighting intensively studied areas (Taupō Volcanic Zone, northern Hikurangi margin, central Alpine Fault) against thinly catalogued offshore tracts, Fiordland, Canterbury, and the far south.

Mature international catalogues of catalogues provide useful reference points: the United States Geological Survey (USGS) Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat) combines the contributing regional and national networks into a single catalogue (USGS, 2025); the Swiss Seismological Service (SED) couples a dense network with cross-border data (Diehl et al., 2025); the European-Mediterranean Seismological Centre (EMSC) and the International Seismological Centre (ISC) aggregate across many agencies (European-Mediterranean Seismological Centre, 2025b; International Seismological Centre, 2025a); and EPOS provides standardised FAIR services across Europe (EPOS, 2025b). These services share common standards, notably the International Federation of Digital Seismograph Networks (FDSN) web services and the QuakeML data format, which New Zealand already supports. Appendix A summarises each of these services. The comparison is a qualitative synthesis from public documentation and expert judgement, not a scored audit; the New Zealand column reflects the authors' assessment across New Zealand's many catalogues, whereas the comparators are single national or federated catalogues of catalogues. Table 2 compares these services across the rows shown; five recurring priorities stand out: (1) consistent representation of uncertainty or confidence; (2) formal version control with Digital Object Identifiers (DOIs) and changelogs for citable releases; (3) a cross-agency event-identifier mapping service; (4) integration of historical and pre-instrumental data; and (5) formal governance with dedicated funding. These gaps motivate the framework components and key recommendations that follow.

Table 2: Comparison of New Zealand and international earthquake-catalogue capabilities. Switzerland (SED) is shown as a detailed national service, and the European federation and aggregation layer (EPOS, EMSC, and ISC) is shown separately. A dash (—) indicates a capability that is not applicable to a single national catalogue or is instead provided at the aggregator level.

Feature	New Zealand	USGS ComCat	Switzerland (SED)	Europe (EPOS / EMSC / ISC)	Gaps & Priorities
Core parameters	Hypocentre (origin time, location, depth), magnitude, phase picks/arrivals, station counts; QuakeML carries location/magnitude uncertainties and multiple magnitudes (GeoNet, 2025a; GNS Science, 2020)	Hypocenter, magnitudes, phase data, moment tensors, macroseismic data (USGS, 2025)	Historical + instrumental events; homogenised M_{Lhc} scale with M_w for notable events; uncertainties, intensities (Diehl et al., 2025)	EMSC/ISC aggregate hypocentres, magnitudes, and phase data from many agency bulletins (International Seismological Centre, 2025a; European-Mediterranean Seismological Centre, 2025b)	Lacks full uncertainty propagation for all events
Metadata & provenance	GeoNet documents core processing; provenance incomplete or inconsistent across the wider NZ catalogue ecosystem	Maintains standards, data dictionaries, method documentation (USGS, 2025)	Documented processing, 3D velocity models, homogenised ECOS database (Diehl et al., 2025; Fäh et al., 2011)	EPOS: standardised metadata and FAIR services across contributing nodes (EPOS, 2025b)	Needs consistency, centralised metadata policies

Continued on next page

Table 2 – *continued*

Feature	New Zealand	USGS ComCat	Switzerland (SED)	Europe (EPOS / EMSC / ISC)	Gaps & Priorities
Versioning & review	Rapid + review workflows; some reprocessing	Versioned updates, multiple origins per event, revisions over time (USGS, 2025)	Static snapshots such as ECOS-09; no per-version catalogue DOIs/changelogs (Diehl et al., 2025)	EPOS: continuous access to evolving, harmonised datasets (EPOS, 2025b)	Formalise version control (DOIs, changelogs)
Access & APIs	GeoNet Quake Search and WFS (CSV, GeoJSON, KML, GML), FDSN (QuakeML), GeoNet API (GeoNet, 2025c,a)	FDSN API, GeoJSON/CSV feeds, search interfaces (U.S. Geological Survey, 2025)	FDSN web services (QuakeML) (Diehl et al., 2025)	EPOS: integrated access to waveforms and parametric data via standards (EPOS, 2025b)	Search across catalogues from one entry point; cross-catalogue queries
Latency / solution status	GeoNet automatic solutions within minutes, then manually reviewed; periodic reprocessing	Real-time GeoJSON feeds; preferred solution revised as reviews land (U.S. Geological Survey, 2025)	Automatic then reviewed solutions (Diehl et al., 2025)	EMSC near-real-time; ISC reviewed bulletin issued ~24 months later (International Seismological Centre, 2025b; European-Mediterranean Seismological Centre, 2025b)	Expose solution status (automatic vs reviewed) and update latency uniformly, per record

Continued on next page

Table 2 – *continued*

Feature	New Zealand	USGS ComCat	Switzerland (SED)	Europe (EPOS / EMSC / ISC)	Gaps & Priorities
Historical data	Some pre-instrumental events with greater uncertainty (GNS Science, 2020)	Includes large historic events, variable completeness (USGS, 2025)	Centuries of data, macroseismic fields, uncertainties (Diehl et al., 2025)	ISC-GEM instrumental reference catalogue for large global events (International Seismological Centre, 2025a)	Strengthen integration, metadata, uncertainty modelling
Coverage & completeness (M_c)	GeoNet $M_c \approx M_L$ 1–2 in well-monitored regions; varies spatially and through time (Warren-Smith et al., 2025; GNS Science, 2020)	$\sim M4.5+$ globally (common analysis threshold); $\sim M2.5$ in well-monitored US regions (USGS, 2025)	M_c 1.0–1.5 via a dense network; record from 1879 (historical to 250 AD) (Diehl et al., 2025; Fäh et al., 2011)	Completeness varies by contributing network and region (International Seismological Centre, 2025a)	Quantify and report completeness (M_c) and coverage consistently across NZ catalogues

Continued on next page

Table 2 – *continued*

Feature	New Zealand	USGS ComCat	Switzerland (SED)	Europe (EPOS / EMSC / ISC)	Gaps & Priorities
Uncertainty & quality	GeoNet reports location/magnitude uncertainties and quality metrics (RMS, azimuthal gap, station/phase counts) via QuakeML; uneven across other NZ catalogues (Warren-Smith et al., 2025; GeoNet, 2025c)	Quality metrics: RMS, station count, azimuthal gap (USGS, 2025)	Emphasises uncertainty and solution quality (Diehl et al., 2025)	EPOS standards for uncertainty and quality (EPOS, 2025b)	Universal uncertainty reporting needed
Multi-catalogue linking	Mostly separate; GeoNet provides limited cross-catalogue mapping by including GCMT moment-tensor solutions for larger (typically offshore) events	Aggregates many networks, links multiple origins per event (USGS, 2025)	— (a single national catalogue)	EPOS harmonises catalogues across Europe; EMSC/ISC reconcile and aggregate bulletins across many agencies (Appendix A) (EPOS, 2025b; International Seismological Centre, 2025a)	Build Event ID mapping service, coordinated portal

Continued on next page

Table 2 – *continued*

Feature	New Zealand	USGS ComCat	Switzerland (SED)	Europe (EPOS / EMSC / ISC)	Gaps & Priorities
Special products	Focal mechanisms, moment tensors, slow slip, LFE as standalone datasets	Hosts moment tensors, additional products tied to events (USGS, 2025)	—	EPOS: integrates fault, slip, and seismological products via standardised services (EPOS, 2025b)	Better integrate special-purpose catalogues
Standards compliance	GeoNet supports FDSN/QuakeML and exports; not all catalogues fully adhere (GeoNet, 2025c)	FDSN and QuakeML in APIs and exports (U.S. Geological Survey, 2025)	Implements FDSN web services; QuakeML (Diehl et al., 2025)	EPOS: builds on FDSN/QuakeML community standards (EPOS, 2025b)	Ensure full compliance across all catalogues
Governance & sustainability	ESNZ/GeoNet or individual researchers; less formal structure	Formal governance, funding, institutional roles (USGS/ANSS/NEIC) (USGS, 2025)	Robust institutional backing and cross-national coordination (Diehl et al., 2025)	EPOS operates as a European Research Infrastructure Consortium (EPOS ERIC) (EPOS, 2025b)	Establish formal governance, roles, funding

3 A Community-Driven Process

The framework proposed in the following sections was shaped through community engagement. Two pre-conference workshops at the Geoscience Society of New Zealand annual meetings provided the primary engagement: a 2023 workshop framed needs and objectives, and a 2024 workshop reviewed progress and refined priorities (Geoscience Society of New Zealand, 2023, 2024). Participants spanned university researchers, GeoNet and Earth Sciences New Zealand staff, government agencies, and engineering and insurance users. Discussion was structured around eight discussion tables (covering minimum catalogue attributes, metadata and documentation, access and versioning, governance, and community priorities), whose outcomes are synthesised as nine cross-cutting themes in Appendix C. Throughout this period, a working group met regularly to draft standards, test integration pathways, and build the prototype portal.

The workshops were designed as scoping and prioritisation exercises rather than as a statistically representative survey or a formal consensus vote. This paper therefore treats their outputs as community-informed evidence for needs and design direction; the proposed standards, governance model, and implementation roadmap are offered for review, calibration, and partner agreement.

3.1 Prototype Portal: Proof of Concept

To ground the framework in a working system, the working group built a functional prototype portal (*Catalogue of Catalogues*, or CofC) for the framework services of Section 4 (incorporation, merge, export, and analytics), using representative New Zealand datasets. Its design drew on the 2023 and 2024 workshops (Geoscience Society of New Zealand, 2023, 2024), and it was first presented, as a conference talk by the authors rather than a workshop with a published page, at the 2025 Geoscience Society of New Zealand annual meeting, where further community feedback informed its development. The prototype is open-source (MIT licence; see Data and Code Availability). It is a proof of concept that exercises the workflows below functionally, establishing their technical feasibility; the national framework of Section 4 builds on that demonstration. The prototype implements:

- **Incorporation** of CSV/tab-delimited text (with automatic delimiter and date detection), JSON/GeoJSON variants, and QuakeML 1.2, with an alias-based field-mapping interface and direct, de-duplicating sync from GeoNet’s FDSN Event service;
- **Search and visualisation:** spatial (bounding box, polygon, radius), time, magnitude, and depth filtering on an interactive map with event clustering, uncertainty ellipses, focal mechanisms, and station coverage;
- **Merge** with five configurable strategies (selection by solution quality, agency, creation time, or completeness, or an averaged solution), conflict detection and logging, lineage tracking, and a preview before any new version is written (Section 4.4);
- **Analytics:** b -value, completeness magnitude (M_c), moment release, declustering, and magnitude-frequency comparison, using standard methods (Aki, 1965; Wiemer and Wyss, 2000; Gardner and Knopoff, 1974); and
- **Multi-format export** (CSV, GeoJSON, KML, QuakeML 1.2), including version-specific exports.

Community feedback from the workshops and the 2025 conference presentation informed refinements to the metadata schema, merge options, and quality-grading approach described in Sections 4.4 and 4.5. The pilot phase (Section 5) is proposed to validate the prototype on named test catalogues, reporting event counts, accepted and rejected records, merge precision and recall,

identifier-matching error rates on dense aftershock sequences, failure cases, and runtime and storage behaviour, with reproduction instructions.

4 The Proposed FAIR Framework

The framework addresses the full handling of catalogue data, from contribution and description through comparison, merging, and citable, permanently retained versions. Figure 3 summarises how the framework’s components are linked to each other. This section describes the framework conceptually. Implementation-level details (API endpoints, access-control roles, field-level representations, and validation thresholds) are collected in Appendix B so that this section stays readable for a general audience.

Unless otherwise stated, the components in this section are proposed design requirements. The discovery-level requirements apply to the submission profile that describes every catalogue and are intentionally lightweight; the service-level requirements apply to catalogues incorporated in full, and especially to DOI-bearing releases, and are correspondingly stricter.

At its core the framework is a coordinating layer, the “catalogue of catalogues”. Its common entry point is a short submission profile (Section 4.2): each catalogue is described by the profile, which records its content and how to obtain it, so that the catalogue is discoverable without necessarily transferring the underlying data. This discovery function is cheap to contribute to and is the immediate community deliverable, answering *which catalogues exist, what they contain, and where to find them*. Building on it, the framework provides richer, FAIR-enabling services [services that make the coordinated catalogues more findable, accessible, interoperable, and reusable (programmatic access, cross-catalogue event-identifier mapping, merge, quality scoring, analytics, and versioned DOIs)] for catalogues that are incorporated in full, answering *compare, merge, analyse, and cite*. A catalogue can therefore be described by a profile that links to where it lives, or opt in to full incorporation for the services that operate on event-level data. These services would be rolled out across the implementation roadmap (Section 5). At the time of writing, the index is a design proposal: it has not yet been populated, and the New Zealand catalogues named in this paper are illustrative examples rather than actual entries. An initial index with an agreed first set of catalogues, via the submission profile, would be the near-term deliverable (Section 5). Table 3 summarises the framework’s two functions, discovery and curation services, before the detailed architecture in Figure 3.

Table 3: Discovery and curation functions of the framework.

Function		Stores	Provides
Discovery		Catalogue-level descriptors (submission profiles) and links to external data sources	Findability (DOI, contact information), broad content flags, and transparent version history of entries
Curation services	ser-	Fully incorporated event-level catalogue data and its processing provenance (the recorded chain of steps, software, and agents, plus merge lineage)	Validation, event-ID mapping, merge, analytics, exports, and DOI-bearing releases

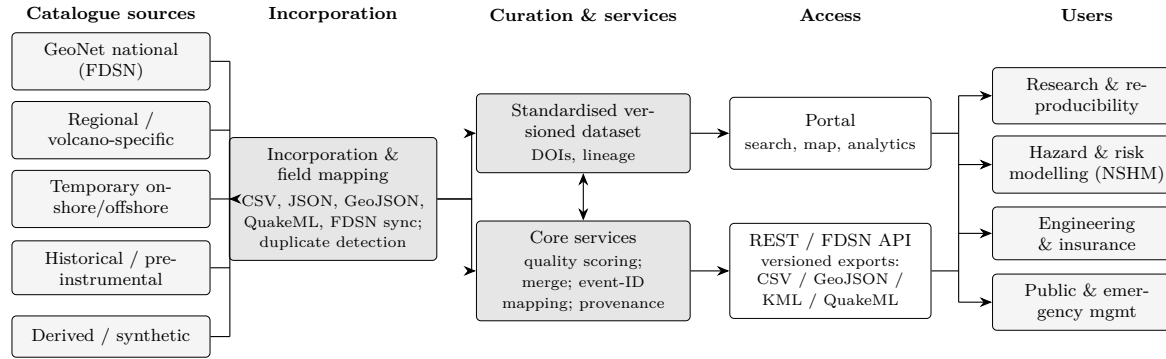


Figure 3: End-to-end architecture of the proposed framework for the full curation services. Heterogeneous catalogue sources selected for full incorporation are mapped to a single, standardised, versioned dataset. Curation services (quality scoring, merge, event-identifier mapping, and provenance) feed a portal and a programmatic interface that serve research, hazard, engineering, and public users. Capabilities identified as implemented in Section 3.1 are exercised functionally by the prototype. The lightweight discovery entry point (the catalogue submission profile, Table 4) is not shown as a separate store here: every catalogue is described by a profile and linked, whether or not it is incorporated in full.

4.1 Metadata Standards and a Shared Lexicon

A core metadata profile aligned with QuakeML and FDSN conventions captures identifiers, origin parameters with uncertainties, magnitude types with errors, detection and quality metrics, methodology descriptors, event classification, and catalogue-level metadata (ETH Zürich Seismology / QuakeML Working Group, 2015; GeoNet, 2025c; U.S. Geological Survey, 2025). Required versus recommended fields are distinguished to balance utility against effort; the full field hierarchy is given in Table 8 (Appendix B). The field-level mapping to QuakeML elements, including how constructs without a native QuakeML element (the composite quality score, cross-catalogue merge lineage, qualitative confidence classes, and macroseismic intensity) are carried via QuakeML’s extension and comment mechanisms, is given in Table 10 (Appendix B). A controlled vocabulary standardises magnitude types, event classifications, phase names and quality codes, units and formats, and dataset naming, improving interpretability and filtering across catalogues (ETH Zürich Seismology / QuakeML Working Group, 2015; Mulder, 2016). An illustrative subset of this lexicon is given in Table 9 (Appendix B).

4.2 Catalogue Submission Profile (Minimum Attributes)

Every catalogue in the framework is described by a concise *submission profile* (Table 4), which includes the minimum set of attributes, which was prioritised at the 2024 community workshop. The profile records, consistently and searchably, *what* a catalogue contains and *how* to obtain it, so that users can discover catalogues and assess relevance before reading a catalogue’s full documentation. Required fields are kept to a small core to lower the barrier to contribution, and controlled drop-downs are preferred over free text wherever possible to keep entries searchable and consistent. Table 4 lists the fields of the submission profile.

Table 4: Catalogue submission profile: minimum attributes for a catalogue-of-catalogues entry, as prioritised at the 2024 workshop. Controlled options are shown in this style; fuller detail is carried by the linked dataset/DOI where one exists.

Field	Status	Options / notes
<i>Identification and contact</i>		
Catalogue name	Required	Short, human-readable title
Contact person(s)	Required	At least one; name and email, multiple separated by semicolons
Data source	Recommended	Origin of the catalogue (e.g. parent catalogue, instruments, records)
Brief description	Required	Why the catalogue was produced and its focus; placed at the end of the form, with a character limit
Additional notes	Optional	Free-text notes/keywords; placed last
<i>Coverage</i>		
Start date	Required	
End date	Optional	Or tick continuous/ongoing
Bounding box	Required	Latitude/longitude (N/S/E/W)
Depth range	Recommended	Minimum and maximum depth
Region of interest	Recommended	Descriptive focus or extent
Magnitude info	Optional	Magnitude range and/or magnitude of completeness, not required (many historical/intensity catalogues lack instrumental magnitudes)
<i>Content present</i> (tick all that apply)		
Content flags	Recommended	hypocentral locations; relative relocations; phase arrival-time picks; amplitude picks; fault-geometry (non-point-source); historical observations (felt reports / macroseismic intensities); quality information; other
<i>Classification</i> (controlled drop-downs)		
Catalogue type	Recommended	background / aftershock / induced / historical / compilation / large-events-only
Focus	Recommended	geographical / parameter-testing / particular-event-type
Detection method	Recommended	manual / automatic / template-matching / machine-learning / felt-reports / other
Review status	Recommended	non-reviewed / partially reviewed / reviewed
<i>Availability</i>		
Data availability	Required	available (DOI) / available on request / not available

Consistent with this scope, the catalogue entry stays high-level and is a discovery record, not a certification of scientific adequacy. Deeper detail is carried by the catalogue’s own dataset or publication and referenced by DOI where one exists, rather than duplicated in the entry. This includes velocity models, station and pick information, processing provenance and parameters, waveform and raw-data links, focal-mechanism details, input-catalogue lineage, links to GeoNet event IDs, and any associated-publication abstract. A graphical, map-based discovery interface (polygon selection and faceted filtering) and contributor self-service editing are planned as future

enhancements; in the interim, catalogue entries are openly searchable and exportable, while catalogues incorporated in full additionally gain the services described below.

4.3 Interoperability: Programmatic Access and Cross-Catalogue Identifiers

A unified API provides machine-readable access across catalogues, returning CSV, JSON, GeoJSON, KML, and QuakeML, with bulk downloads of versioned datasets to support reproducible research; it builds on existing GeoNet interfaces and standards (GeoNet, 2025d,c; ETH Zürich Seismology / QuakeML Working Group, 2015). The endpoint design is given in Appendix B.

A persistent interoperability challenge is that the same earthquake carries different public identifiers across agencies (a same New Zealand event may have a GeoNet Public ID, a USGS event code, and an ISC event number), so researchers aggregating sources must de-duplicate by hand. The framework includes an event-identifier mapping service that maintains cross-agency equivalences, matched using configurable spatial, temporal, and magnitude thresholds (illustratively, epicentral separation < 25 km, time offset < 30 s, magnitude difference < 0.5 , depth difference < 20 km). For comparison, the ANSS composite catalogue identifies duplicates on space and time alone, using operational windows of 16 s and 100 km (Northern California Earthquake Data Center, 2024); the magnitude criterion is added here to reduce false matches in dense sequences. These default thresholds would be calibrated during the pilot phase against labelled New Zealand event pairs, dense aftershock sequences, and cross-agency catalogues to estimate false-match and missed-match rates. Because wide windows can group distinct events in dense sequences, thresholds are configurable per source and ambiguous matches are flagged for expert review rather than merged automatically. The service aligns with the approach used by EMSC (European-Mediterranean Seismological Centre, 2025b) and extends it to the New Zealand context.

4.4 Catalogue Merge

Catalogue merging, reconciling the multiple solutions that different catalogues report for the *same* earthquake into a single, unified view, is a central operation that builds on established cross-agency practice (Si et al., 2024; Northern California Earthquake Data Center, 2024). We use *merge* for this per-event reconciliation, as distinct from *aggregation*: assembling many catalogues or agency bulletins into one collection, in the sense of the ISC and EMSC, without necessarily reducing the solutions held for each event.¹ The framework specifies five merge strategies, all exercised by the prototype (Section 3.1); users choose by scientific objective:

- **Quality-based** (recommended): the best-scoring solution is retained. The prototype ranks candidates with a coarser merge-time solution score (station count, azimuthal gap, RMS, magnitude uncertainty and type, evaluation status); ranking on the full quality score of Section 4.5 is specified for full implementation.
- **Priority-based**: a nominated authoritative agency’s solution is preferred; others are recorded as alternatives.
- **Newest data**: the most recently determined solution is retained, where later analyses supersede earlier ones.
- **Most complete**: the solution with the most populated fields is retained, maximising metadata density.

¹A companion review of catalogue-merging strategies accompanies this paper in the project repository (see Data and Code Availability).

- **Averaged:** location, depth, and magnitude are combined across solutions by inverse-variance weighting, using an ISC magnitude-type hierarchy (a preferred ordering of magnitude types - for example M_w ahead of M_L - that avoids scale saturation). Agency solutions for one event often share data and processing assumptions, so this strategy is best treated as exploratory unless validated for the catalogues being combined. Its merged uncertainty is reported conservatively (no smaller than the smallest contributing value) rather than the tighter value that would follow from assuming the solutions are statistically independent.

These five strategies each yield a single preferred solution per event. As an alternative, the framework can instead *aggregate* all contributing solutions under one event, following the QuakeML data model in which an event may carry several origins and magnitudes with one marked as preferred, as in the ISC and EMSC bulletins (International Seismological Centre, 2025a; European-Mediterranean Seismological Centre, 2025b). Each retained origin keeps its own quality grade (Section 4.5), so users can compare solution qualities within an event and defer the choice of preferred solution to a later stage, at the cost of a larger, less reduced record.

For the averaged strategy, a quantity x reported by sources i with standard deviations σ_i is combined as the inverse-variance-weighted mean, alongside the variance that would follow *if* the solutions were statistically independent:

$$\bar{x} = \frac{\sum_i w_i x_i}{\sum_i w_i}, \quad w_i = \frac{1}{\sigma_i^2}, \quad \sigma_{\text{indep}}^2 = \frac{1}{\sum_i w_i}.$$

The weights w_i pull the mean toward the better-constrained solution. However, agency solutions for the same event typically share waveforms, phase picks, and processing assumptions, so they are correlated rather than independent, and σ_{indep} understates the true uncertainty. The framework therefore reports a conservative uncertainty floored at the most precise contributing solution,

$$\sigma_{\text{reported}} = \max(\sigma_{\text{indep}}, \min_i \sigma_i).$$

Two further realities constrain when averaging is valid. First, uncertainties are reported inconsistently across catalogues, as different quantities and often with fields left empty, so a solution that lacks a comparable quantified uncertainty is excluded from the inverse-variance weighting rather than assigned a placeholder value. Second, uncertainties from double-difference relocations (for example GrowClust or hypoDD) are relative to a cluster centroid rather than absolute, and are therefore systematically smaller; because the weights scale as $1/\sigma_i^2$, including such a solution at face value would bias the weighted mean toward it. The framework flags relocation catalogues of this kind and, by default, excludes their relative uncertainties from the weighting (or rescales them to an absolute basis) and warns the user. The averaged strategy is intended for exploratory use and emits this warning whenever solutions of mixed uncertainty provenance are combined.

This is the framework’s specified averaged-merge behaviour. The prototype currently realises the epicentre inverse-variance averaging and, to avoid magnitude-scale saturation, selects magnitude by the ISC type hierarchy and depth from the lowest-uncertainty solution; inverse-variance depth weighting and the conservative merged-uncertainty floor are specified for full implementation.

Conflicts are detected and logged when solutions disagree beyond configurable thresholds (spatial separation, temporal offset, magnitude difference, depth range, network mismatch); duplicate detection during dense aftershock sequences follows nearest-neighbour methods (Vorobieva et al., 2022). Each merged event retains explicit lineage to its source events and the configuration used, enabling full audit and reproducibility; a preview reports conflict statistics without writing a new version. Production merge configurations should report validation results and sensitivity to threshold choice. Table 5 works this through for one event reported by two catalogues, and Figure 4 shows the resulting epicentre.

Table 5: Synthetic worked example of the *averaged* merge strategy for one earthquake reported by two catalogues (Source A is entry `catA-0001`, Source B is `catB-0172`; the merged event `merged-0001` retains lineage to both). Each quantity is the inverse-variance-weighted mean of the two solutions; the normalised weights $w_A:w_B$ (with $w_i = 1/\sigma_i^2$) show how the mean is pulled toward the better-constrained Source A. The last column contrasts the uncertainty obtained by *incorrectly* assuming the solutions are independent, σ_{indep} , with the conservative value actually reported, $\sigma_{\text{reported}} = \max(\sigma_{\text{indep}}, \min_i \sigma_i)$ (Section 4.4); the floor is active in every row. Latitude and longitude are weighted by the horizontal uncertainty, so their merged epicentre uncertainty is the *Horizontal unc.* row. Values and identifiers are illustrative and are not official event solutions.

Quantity	Source A	Source B	Weights $w_A:w_B$	Merged mean	Merged σ (in- dep. $\rightarrow$ rep.)
Origin time (s)	12.3 ± 0.5	12.7 ± 0.8	0.72 : 0.28	12.4	$0.42 \rightarrow 0.50$
Latitude ($^\circ$)	-38.520	-38.524	0.80 : 0.20	-38.521	n/a
Longitude ($^\circ$)	176.120	176.150	0.80 : 0.20	176.126	n/a
Horizontal (km)	unc. 1.5	3.0	n/a	n/a	$1.34 \rightarrow 1.50$
Depth (km)	8.0 ± 2.0	9.0 ± 4.0	0.80 : 0.20	8.2	$1.79 \rightarrow 2.00$
Magnitude (M_w)	4.2 ± 0.1	4.3 ± 0.2	0.80 : 0.20	4.22	$0.089 \rightarrow 0.10$

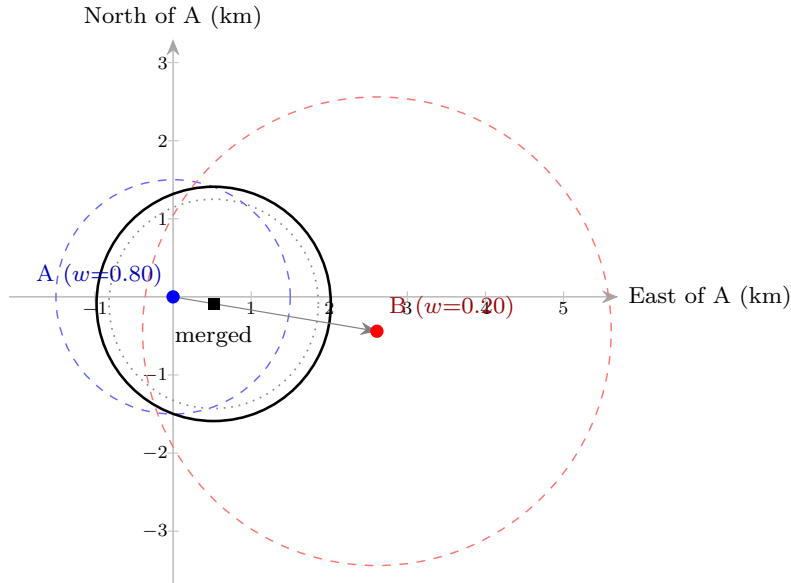


Figure 4: The averaged merge of Table 5, in a local East–North frame centred on Source A. The inverse-variance weights (0.80 : 0.20 horizontally) pull the merged epicentre (black square) about one-fifth of the way from the better-constrained Source A (blue) toward Source B (red); the dashed circles are the 1σ horizontal uncertainties (1.5 and 3.0 km). The merged uncertainty is *not* allowed to shrink below the most precise input: the solid circle is the reported $\sigma = 1.5$ km, whereas the dotted circle is the smaller $\sigma_{\text{indep}} = 1.34$ km that the (invalid) independence assumption would give. Epicentre coordinates are plotted from the unrounded weighted mean.

4.5 Data Quality Assessment

A standardised, transparent quality score can support informed reuse decisions, building on established practice for assessing catalogue quality (Woessner and Wiemer, 2005) and on the

criteria used to benchmark GeoNet location quality (Warren-Smith et al., 2025). Each event receives a score (0–100) computed as a weighted sum over five dimensions of solution quality:

- **Location** (35 %): horizontal, depth, and time uncertainties, weighted most heavily on horizontal uncertainty;
- **Network geometry** (25 %): azimuthal gap and used-station and used-phase counts, the primary instrumental predictors of location error (Bondár et al., 2004; Warren-Smith et al., 2025);
- **Solution fit** (15 %): the RMS arrival-time residual of the location;
- **Magnitude** (15 %): magnitude uncertainty and the number of contributing stations; and
- **Evaluation** (10 %): manually reviewed/final solutions score above automatic/preliminary ones.

Each dimension penalises missing fields, so sparsely documented events are scored conservatively. Field completeness, physical-consistency, and magnitude-homogeneity checks are handled separately by the automated validation suite (Appendix B). The score maps to the grade bands in Table 6. The weights and bands are configurable defaults; the pilot phase will calibrate them against representative New Zealand catalogues. The grades are screening guidance based on documentation and solution indicators, not a measure of event accuracy or of fitness for a specific hazard, engineering, or scientific analysis. Flagged statistical outliers are annotated for caution rather than removed automatically.

Table 6: Quality grade bands and screening guidance (default scheme; the pilot phase will calibrate the weights and bands; full seven-band scheme in Appendix B). The dimensions and thresholds underlying the score follow established practice for assessing catalogue quality (Woessner and Wiemer, 2005), the network-geometry predictors of location accuracy (Bondár et al., 2004), and the criteria used to benchmark GeoNet location quality (Warren-Smith et al., 2025); the band boundaries themselves are a design choice, not derived from those studies.

Grade	Score	Interpretation and screening guidance
A	85–100	Strong solution indicators; still verify suitability for the intended analysis.
B	65–84	Usable for screening; document limitations or supplement metadata before quantitative use.
C	45–64	Weak indicators; expert review recommended before quantitative use.
D/F	<45	Minimal documented support; discovery or illustrative purposes only.

4.6 Versioning, Persistent Identifiers, and Provenance

Every catalogue release carries a version, timestamp, and changelog, and all versions remain accessible for reproducibility. The framework uses a three-level MAJOR.MINOR.PATCH scheme (worked examples in Table 11, Appendix B): *major* increments alter event parameters, *minor* increments add events or optional fields, and *patch* increments change only metadata or documentation.

Persistent identifiers make those releases citable. The catalogue series carries a single concept DOI, and each major and minor release (the changes that affect citable content) receives its own version DOI (DataCite, 2025); patch releases update the existing record without a new DOI, and

frequent minor releases may be batched (for example, quarterly) to avoid identifier proliferation. A machine-readable changelog accompanies every release, so a researcher can cite the exact data state used in a study and a reviewer can see what changed between versions.

Provenance, the documented processing history of a solution, is recorded following the W3C PROV model of entities, activities, and agents: each processing step (detection, picking, location, velocity-model choice, magnitude calibration) is captured, with software version and execution date as minimum requirements (World Wide Web Consortium, 2013). This provenance is richer than the per-origin **agencyID** or a single source DOI, capturing how a solution was produced rather than only who produced it. Uncertainties for time, location, and magnitude are reported as standard deviations (with confidence-ellipse parameters where anisotropic) and always carry the magnitude type code; events without quantified uncertainties are flagged, and historical events may carry a qualitative confidence class (e.g. “well-constrained”, “felt-only”). Field-level representations are specified in Appendix B.

4.7 Special Catalogue Types

Different catalogue types need adapted handling to preserve scientific value. Regional research catalogues, in particular, are incorporated as distinct, separately versioned catalogues and cross-referenced (not silently merged) with the national catalogue, because their differing processing assumptions can introduce systematic biases; temporary deployments, historical and pre-instrumental catalogues, and derived or synthetic catalogues likewise receive type-specific treatment, detailed in Appendix B.

4.8 Licensing and Access; FAIR Compliance Indicators

To support FAIR reusability, every dataset and metadata record should carry an explicit, machine-readable licence (Wilkinson et al., 2016; GO FAIR Initiative, 2025). For government-funded baseline products, a default open licence is recommended, aligned with New Zealand Government Open Access and Licensing (NZGOAL) guidance (New Zealand Government, 2022), with documented exceptions for embargoed, sensitive, or culturally restricted data. Access is governed by a role-based model with catalogue-level public/registered/restricted designations and, where needed, field-level masking of culturally sensitive locations; all write operations are audited. These mechanisms are detailed in Appendix B. Progress should be tracked against a FAIR compliance matrix that maps each FAIR sub-principle (F1–F4, A1–A2, I1–I3, R1.1–R1.3) to a framework mechanism and a measurable, annually reported indicator: for example, the percentage of catalogues with a resolvable DOI (F1), mean metadata-completeness (F2), API uptime (A1), and the percentage carrying a machine-readable licence (R1.1). The full matrix is given in Table 14 (Appendix B).

4.9 Aotearoa New Zealand Data Governance: FAIR with CARE

In Aotearoa New Zealand, openness operates within Te Tiriti o Waitangi: the Crown–Māori partnership creates obligations of governance, protection, and active participation that extend to data about, or of significance to, tangata whenua. Earth Sciences New Zealand is a Crown research entity, so the framework must give effect to those obligations, applying the CARE Principles for Indigenous Data Governance (Carroll et al., 2020) alongside FAIR and consistent with Māori data-sovereignty practice (Kukutai and Taylor, 2016).

Concretely, the four CARE principles are operationalised as follows. *Collective benefit*: engagement and benefit-sharing arrangements for catalogues of significance to iwi and hapū. *Authority*

to control: iwi and hapū determine the access and use conditions for their data; the portal’s access controls (Appendix B) only *implement* conditions that iwi and hapū set, and never substitute for that authority. *Responsibility*: documented engagement pathways, attribution, and reporting back to communities. *Ethics*: review of culturally sensitive records before release. Here “mātauranga-informed” means catalogue records or interpretations that draw on Māori knowledge, for example, oral-tradition accounts of historical earthquakes or place-based knowledge of shaking and ground effects, which carry their own provenance and access expectations.

For this commitment to be substantive rather than aspirational, iwi and hapū must be engaged as partners and decision-makers from the outset, and the Steering Committee (Section 5.1.1) must include Māori membership with a substantive governance role, not a single advisory seat. Phase 1 will define, with iwi and hapū partners, how culturally significant records are identified, who has authority to set access conditions, how representatives are selected and resourced, and how disputes over access, attribution, or release are resolved. The CARE implementation and access-control model will be co-developed and reviewed with mandated Māori data-governance partners.

5 Implementation: Governance, Roadmap, and Funding

Sustainable implementation will require governance arrangements that support long-term stewardship, community participation, and alignment with both FAIR and CARE principles. The governance model described below is proposed as one possible approach for community consideration and refinement.

5.1 Governance, Roles, and Responsibilities

5.1.1 Steering Committee

A Steering Committee is proposed to provide strategic oversight and ensures community ownership. Membership could comprise representatives from Earth Sciences New Zealand, at least two universities with active seismology programmes, a government user (e.g. the National Emergency Management Agency), an engineering or insurance representative, and, where feasible, an international advisor from a peer organisation (e.g. ISC, EMSC, or the GFZ Helmholtz Centre for Geosciences). To support alignment with CARE principles and Māori data interest, membership could include Māori representation with the mandate and resourcing to speak to data-sovereignty matters (see Section 4.9). The process for appointing Māori members could be developed with iwi and hapū partners rather than set unilaterally by the host institution.

The Steering Committee is proposed to meet at least quarterly with defined decision-making processes and staggered three-year terms. The Committee may approve new data sources for incorporation, ratify versioning decisions affecting the national record, oversee international partnerships (e.g. alignment with EPOS), and commission annual reporting against FAIR compliance indicators.

5.1.2 Data Stewardship Team

Routine operation is proposed to be led by Earth Sciences New Zealand and need not be large. In its minimal configuration, the framework is proposed to be maintained by a part-time data steward (coordination, metadata curation, and quality review); potential resourcing and FTE scaling are set out in Section 5 (Funding and Resourcing). External researchers may also

submit catalogues through a defined contribution process (Section 5.1.3). These governance roles (contributor, reviewer, maintainer) are proposed to be distinct from the portal’s technical access roles defined in Appendix B.

5.1.3 Catalogue Contribution Process

External researchers and institutions may submit catalogues for inclusion. Submissions are proposed to require: catalogue name and description, responsible institution and contact, time period and geographic coverage, event count where known, format and metadata documentation, and a declared licence; magnitude range may be required where instrumental magnitudes are part of the catalogue and otherwise recorded as not applicable. A two-stage review process is proposed: technical QC by the Data Stewardship Team (format validity, completeness score, duplicate check) followed by scientific peer review by a standing panel of subject-matter seismologists who are independent of the submitted catalogue where practical (methods assessment, comparison with existing catalogues). Because the Committee is proposed to meet only quarterly, responsibility may be divided to keep the contribution turnaround within a target of four to eight weeks: the Steering Committee is proposed to set incorporation policy and approve *categories* of source, while the Team may approve individual contributions against that policy. Only novel or contentious cases may be escalated to the Committee. Written feedback is proposed to be provided regardless of outcome, and approved catalogues are proposed to be assigned a DOI and version record.

5.1.4 Catalogue Custody, Updates, and Use Conditions

Each catalogue entry could name a custodian (the person responsible for it) with a listed contact for updates, corrections, or withdrawal. The custodian would not need not be the original author, so that past studies and legacy datasets can be entered by others; in such cases the entry could attribute the catalogue’s authorship. Where the same catalogue might be entered more than once (for example, a legacy dataset entered independently), duplicate entries could be identified and reconciled by the Data Stewardship Team. The catalogue index itself is proposed to be version-controlled, with its release history maintained in a public Git repository, so that additions, edits, and deprecations would be transparent and a superseding catalogue version could be linked to the entry it replaces.

The catalogue index would be a finding aid, not an endorsement (the submission profile itself is defined in Section 4.2). A clear disclaimer could state that entries are contributed and not independently verified, and that the index would simply records which catalogues exist. Users could be encouraged to reference catalogues appropriately and to contact and collaborate with the original authors where relevant, consistent with the intellectual-property and data-sharing expectations attached to each dataset.

5.1.5 Success Metrics and Annual Reporting

Successful implementation does not require the replacement of existing catalogues, the centralisation of ownership, or the adoption of a single catalogue format. Rather, success will be achieved when earthquake catalogues can be readily discovered, understood, cited, compared, and reused through shared metadata, transparent provenance, and interoperable services, while stewardship remains with the appropriate custodians.

Progress could be measured annually through a combination of operational, scientific and community indicators. Measures may include the proportion of known New Zealand earthquake catalogues within the framework; participation by catalogue custodians and organisations;

metadata completeness and FAIR compliance; usage metrics such as registered users, downloads, and API queries; and infrastructure performance measures including service availability and reliability. Reported measures include: the number of catalogues listed and incorporated; total event count and geographic coverage; metadata-completeness distributions; API uptime (target 99.5%); registered users; downloads and API queries;

Scientific and user-focused indicators are equally important. These may include peer-reviewed publications, hazard assessments, engineering studies and other applications that cite versioned catalogue releases; reproducibility checks on cited versions, and user-reported time; benchmarking of catalogue merge performance on representative datasets; and user-reported reductions in effort, duplication, or error when working across multiple catalogues .

Reporting against the FAIR sub-principle indicators (Table 14) could also be included, and annual reports published openly.

To provide a sense of scale, indicative two-year milestones could include the GeoNet national catalogue and at least 20 or more additional regional, temporary, and historical catalogues being discoverable through the framework; , with priority datasets incorporated for the full services; 200 or more registered users across the research, engineering, and agency communities; 10 or more research outputs citing a versioned catalogue DOI; and API availability approaching the 99.5% operational target. Baselines, targets and measurement methods should be refined during the initial implementation phase in consultation with participating organisations and users.

5.2 Roadmap

Development proceeds in four phases (Figure 5): planning and mobilisation; prototyping and pilot integration; full-scale implementation and public launch; and sustained operation as a national core service with annual review and FAIR compliance reporting.

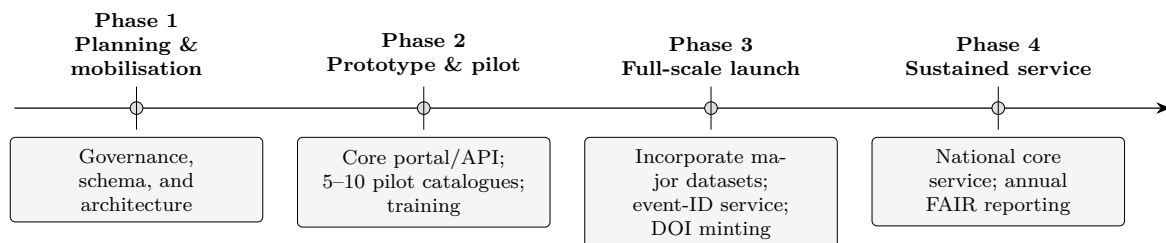


Figure 5: Indicative four-phase roadmap from planning to a sustained national service.

5.3 Funding and Resourcing

Indicative resourcing is modest, and the discovery function in particular is inexpensive to run. It can be launched and maintained by a part-time data-stewardship role (coordination, metadata curation, and periodic review) with low-cost hosting for a versioned index (Section 5.1.2), on the order of a few tens of thousands of dollars per year. A mature service that also operates the full curation and incorporation services would scale to perhaps one to two FTE, adding software-engineering and user-support effort, together with modest cloud infrastructure and recurring DOI-registration fees. A realistic path therefore begins small, with discovery, and grows only as the full services are built out and demand justifies it. Indefinite retention of every versioned release is a growing-storage liability; it is bounded by keeping only changelogs and metadata for superseded *patch* releases, retaining full event tables only for DOI-bearing major and minor versions, and moving older releases to tiered/cold storage. A natural home is a national

research-infrastructure line (for example, an Earth Sciences New Zealand core-function budget or a research-infrastructure programme such as those administered by the Ministry of Business, Innovation and Employment, MBIE), so that the service is funded as a standing capability rather than a fixed-term project. The figures here are indicative of scale, not costed estimates; precise budgets should be developed with the host institution.

6 Benefits by Stakeholder

A coordinated, FAIR-and-CARE catalogue capability delivers concrete, audience-specific value:

- **Researchers** gain discoverable, citable, version-pinned datasets and ready cross-catalogue comparison, making studies faster to run and easier to reproduce.
- **Hazard and risk modellers** (e.g. the National Seismic Hazard Model) get comparable, documented inputs with consistent magnitudes and uncertainties, reducing the bespoke harmonisation that the 2022 NSHM required (Gerstenberger et al., 2024; Christophersen et al., 2024).
- **Engineering and insurance** users obtain traceable inputs with explicit licences and transparent quality scores, supporting more defensible design and risk decisions.
- **Emergency management and the public** benefit from clearer, better-documented event information and a single national point of discovery, while GeoNet remains the authoritative operational source.
- **Iwi and hapū** should be engaged as partners in governance, setting the access and attribution conditions for data of significance to them, which the framework then implements (Section 4.9).
- **The national system** preserves legacy and temporary datasets that would otherwise be at risk of loss, and aligns New Zealand with international practice (EPOS, 2025b).
- **Community infrastructure** gains shared infrastructure, the catalogue of catalogues, that supports broader community goals identified at the workshops: digitising and back-filling legacy catalogues, and converging on a national community velocity model (1D and 3D). The framework should accept routine updates, with its governance and schema reviewed on a regular cadence.

7 Risks and Limitations

The framework is intended as a practical and incremental approach to improving the discoverability, interoperability, and stewardship of New Zealand’s earthquake catalogues. Nonetheless, a number of risks, limitations, and commonly raised concerns should be acknowledged and considered during implementation.

- **Relationship to GeoNet.** The framework does not duplicate GeoNet; it is an aggregation and curation layer over GeoNet and the catalogues outside it, and GeoNet remains the authoritative operational source and a primary data provider (Figure 3).
- **Long-term funding.** The service is designed as a standing national capability hosted on a core-function or research-infrastructure budget rather than a fixed-term project. The indicative steady-state cost and a bounded-retention storage policy are given in Section 5 (Funding).

- **Contribution incentives.** The contribution process is low-friction and offers tangible incentives (DOIs, citations, increased visibility, and quality feedback), and the system provides value from GeoNet and pilot catalogues from the outset.
- **Relation to existing international systems.** EPOS, ISC, and ComCat are complementary, and the framework is designed to interoperate with them; none, however, curates New Zealand’s regional, temporary, and historical catalogues or addresses CARE obligations, so a national capability remains necessary.
- **Preservation of scientific differences.** Standardisation does not flatten legitimate scientific differences: alternative solutions are preserved with lineage, regional catalogues are cross-referenced rather than silently merged, and merge strategies are explicit and reproducible.
- **Variable metadata quality and completeness.** Many catalogues were developed for specific research purposes and may not contain the metadata, uncertainty estimates, provenance information, or documentation required for full FAIR compliance. The proposed framework is therefore designed to accommodate varying levels of metadata completeness through a minimum submission profile, transparent quality assessment, and progressive improvement over time rather than requiring immediate standardisation of all legacy catalogues.
- **Community adoption.** The value of the framework depends on participation by catalogue custodians, data providers, and users. While the proposed approach seeks to minimise barriers through a lightweight submission process and incentives such as increased visibility, persistent identifiers, and citation opportunities, participation is expected to occur gradually and may vary across catalogue types and organisations.
- **Scope and expectations.** The framework is intended to improve discoverability, interoperability, provenance, and reuse. It is not intended to replace specialist expertise, resolve all scientific disagreements, or guarantee uniform catalogue quality. Users remain responsible for assessing the suitability of a catalogue for their intended application.

8 Conclusion and Call to Action

Aotearoa New Zealand has developed a rich and diverse earthquake catalogue ecosystem through decades of public investment in earthquake monitoring, scientific research, hazard assessment, and temporary seismic deployments. The data, expertise, and technical capability required to make this ecosystem FAIR already exist; what is currently absent is a coordinating framework and the governance to sustain it over the long term. This white paper has set out such a framework, compared it with international practice, and demonstrated its technical feasibility through a working prototype. The remaining step is collective review, partner agreement, and sustained implementation.

The following steps are proposed for GeoNet and Earth Sciences New Zealand, universities, government agencies, iwi and hapū partners, and the engineering and insurance sectors:

1. review, refine, and, where appropriate, endorse the metadata, lexicon, versioning, and quality-scoring standards proposed here as a shared national baseline;
2. stand up the “catalogue of catalogues” as the immediate, low-cost first step, beginning with discovery (populated via the agreed submission profile) ahead of building out the full FAIR services;

3. consider and refine governance arrangements, including the proposed Steering Committee and Data Stewardship Team, to support implementation in a manner consistent with both FAIR and CARE principles (Sections 5.1.1 and 5.1.2), with representation that operationalises both FAIR and CARE; and
4. support a multi-year funding case to move from prototype to a sustained national service.

Collectively, New Zealand’s earthquake catalogues represent decades of scientific effort, public investment, and observational infrastructure. Their long-term value depends not only on their continued creation, but also on their discoverability, interoperability, and stewardship. This white paper presents a community-informed pathway towards achieving that vision and invites the earthquake science, engineering, and hazard communities to contribute to its development and implementation.

Data and Code Availability

In keeping with the FAIR principles it advocates, the prototype portal source code, the metadata schema, the controlled parameter lexicon, the companion review of catalogue-merging strategies, and a companion inventory of New Zealand earthquake catalogues (1960–2026) are available under the MIT licence at <https://github.com/KennyGraham1/catalogofcatalogs> (documentation at <https://catalogofcatalogs.readthedocs.io/>); consistent with the persistent-identifier practice this paper advocates, a citable DOI for a versioned snapshot of these artefacts will be minted (e.g. via Zenodo) with the first tagged release. The earthquake catalogues used as examples are available from their respective providers: the GeoNet national catalogue (GeoNet, 2025a), USGS ComCat (USGS, 2025), and the Swiss Seismological Service (Diehl et al., 2025).

Acknowledgements

We thank participants in the 2023 and 2024 workshops and at the 2025 Geoscience Society of New Zealand annual meeting for their contributions, and colleagues across Earth Sciences New Zealand and partner universities for their input to this framework.

A International Benchmark Detail

This appendix expands the summary in Section 2 and Table 2.

A.1 New Zealand (GeoNet)

GeoNet maintains New Zealand’s earthquake catalogue as part of the National Earthquake Information Database, the central repository for NZ seismic events (GNS Science, 2020). The catalogue contains origin time, location, magnitude, and seismic phase data, with each event assigned a unique Public ID (GeoNet, 2025a), spanning instrumentally recorded earthquakes (1930s–present) and historical events from oral/written records (GNS Science, 2020). Data access includes Quake Search (CSV, GeoJSON, KML, GML) (GeoNet, 2025a) and services compliant with the International Federation of Digital Seismograph Networks (FDSN) and Web Feature Service (WFS) standards, delivering QuakeML with uncertainties, phase arrivals, and multiple magnitude determinations (GeoNet, 2025c,b). Completeness is approximately M_L 1–2 in well-monitored regions (higher elsewhere and through time); location quality varies markedly

with station coverage, as quantified by Warren-Smith et al. (2025). A 2022 revision moved the catalogue toward more consistent magnitude reporting nationwide (GNS Science, 2020).

A.2 United States (USGS ComCat)

The United States Geological Survey (USGS) operates the federated Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat), covering global earthquakes ($\sim M4.5$ and above as a common global analysis threshold, and roughly $M2.5$ in well-monitored US regions) (USGS, 2025), accessed via the FDSN Event API with QuakeML, CSV/GeoJSON/KML, and real-time GeoJSON feeds (U.S. Geological Survey, 2025). Entries include quality metrics (stations used, azimuthal gap, RMS residuals) and magnitude uncertainty; ComCat integrates the ANSS regional and national networks, stores multiple solutions per event with one preferred, and hosts products such as ShakeMap and crowdsourced intensity data (USGS, 2025).

A.3 Switzerland (SED)

The Swiss Seismological Service (SED) maintains a high-quality catalogue dating to 1879 (Diehl et al., 2025), with historical events to 250 AD. A dense network achieves M_c 1.0–1.5 completeness; SED introduced updated magnitude scales (M_{Lhc} , 2020), employs 3D velocity models and relative relocation, and provides formal uncertainties via QuakeML since 2013 (Diehl et al., 2025). Data are accessible through FDSN web services, and cross-border collaboration integrates 60+ stations from France, Germany, Italy, and Austria (Diehl et al., 2025). The ECOS project unified historical records into a homogenised database (Fäh et al., 2011).

A.4 European and Global Aggregators

The European-Mediterranean Seismological Centre (EMSC) operates the European Seismic Portal, aggregating data from multiple networks via FDSN-compliant services and providing multiple solutions and phase data (European-Mediterranean Seismological Centre, 2025a,b). The International Seismological Centre (ISC) archives global bulletin data from 130+ agencies, accepting multiple formats (Nordic, GSE, ISF, QuakeML) (International Seismological Centre, 2025a) with output in ISF or QuakeML (International Seismological Centre, 2025c). Other agencies (the Japan Meteorological Agency, Geoscience Australia, and the European Integrated Data Archive) have adopted FDSN services, and major catalogues now share common standards for open, interoperable access (EPOS, 2025b; ETH Zürich Seismology / QuakeML Working Group, 2015).

B Technical Specification

This appendix collects the implementation-level detail referenced from Section 4. It defines the reference design; specific field representations and thresholds may be refined with partners during Phase 1.

B.1 Metadata Field Hierarchy

Table 7 separates requirement levels by catalogue scope, and Table 8 gives the catalogue- and event-level metadata profile for incorporated records.

Table 7: Metadata requirement levels across the framework.

Scope	Requirement level	Purpose
Discovery entry (submission profile)	Minimal descriptor required	Discovery, attribution, contact, coverage, and availability; data may remain external or unavailable.
Incorporated catalogue	Event-level structure required where applicable	Enables validation, search, analytics, exports, and event-ID mapping.
DOI-bearing release	Version, changelog, licence, and persistent identifier required	Ensures a citable, reproducible data state.
Legacy or historical record	Missing quantified fields must be flagged	Preserves discoverability while avoiding false precision.

Table 8: Metadata Field Hierarchy for Incorporated Catalogue Records

Field	Status	Description
<i>Catalogue-level</i>		
Persistent identifier	Required for DOI-bearing releases	DOI or equivalent resolvable identifier; discovery-only entries may instead point to an external contact or source
Name, description	Required	Human-readable title and summary
Institution, contact	Required	Responsible party and contact details
Temporal coverage	Required	Start and end date of events
Geographic coverage	Required	Bounding box or region descriptor
Magnitude range	Required where applicable	Minimum and maximum magnitude; record as not applicable for catalogues without instrumental magnitudes
Event count	Required where known; required for incorporated catalogues	Total number of events
Licence	Required	Machine-readable licence identifier (e.g. CC-BY 4.0)
Processing summary	Required	High-level description of methods
Velocity model	Recommended	Reference to velocity model used
Location algorithm	Recommended	Software and version (e.g. NonLinLoc 7.0)
Magnitude calibration	Recommended	Scale definitions and calibration reference
Publication links	Recommended	DOIs of associated papers
<i>Event-level</i>		
Origin time	Required	UTC time (ISO 8601)
Latitude, longitude	Required	WGS84 decimal degrees
Depth	Required	Kilometres below sea level
Magnitude value & type	Required where applicable	Value and scale code (e.g. M_w , M_L); historical/intensity-only records may instead carry intensity or qualitative size fields
Origin time uncertainty	Required where available	Seconds (1σ); otherwise flag absence explicitly
Horizontal uncertainty	Required where available	Kilometres (1σ); otherwise flag absence explicitly
Vertical uncertainty	Required where available	Kilometres (1σ); otherwise flag absence explicitly
Magnitude uncertainty	Required where available	Standard deviation; otherwise flag absence explicitly
Station count	Recommended	Number of stations used
Azimuthal gap	Recommended	Degrees
Phase count	Recommended	Number of phase picks used
RMS residual	Recommended	Seconds
Evaluation status	Recommended	“automatic” or “reviewed”
Event type	Recommended	QuakeML event type vocabulary

B.2 Controlled Parameter Lexicon

Table 9 shows representative entries from the controlled lexicon referenced in Section 4. It is an illustrative subset, aligned with the QuakeML enumerations (ETH Zürich Seismology / QuakeML Working Group, 2015; Mulder, 2016), not the complete normative list. A fuller controlled vocabulary is maintained with the schema in the project repository (see Data and Code Availability); its normative, versioned form will be published with the first DOI-bearing release.

Table 9: Illustrative subset of the controlled parameter lexicon (representative terms; aligned with QuakeML).

Category	Representative controlled terms
Magnitude type	Mw (moment); ML, MLv (local); mb (body-wave); Ms (surface-wave); Md (duration)
Event type	earthquake; induced or triggered event; quarry blast; explosion; volcanic eruption; other event
Phase name	P, S; Pn, Pg, Sn, Sg (crustal); pP, sP (depth phases)
Evaluation mode / status	mode: manual / automatic; status: preliminary / confirmed / reviewed / final / rejected
Pick onset / polarity	onset: emergent / impulsive / questionable; polarity: positive / negative / undecidable
Depth determination	from location; operator assigned; constrained by depth phases; constrained by direct phases
Units & formats	time: ISO 8601 UTC; coordinates: WGS84 decimal degrees; depth: km below sea level

B.3 QuakeML/FDSN Crosswalk

The framework adopts QuakeML 1.2 and the FDSN web-service conventions as its interchange backbone (ETH Zürich Seismology / QuakeML Working Group, 2015). Table 10 maps the framework’s catalogue constructs to specific QuakeML elements. Most map directly to native QuakeML. QuakeML also permits free-text `comment` elements and custom elements from a foreign namespace; the framework uses these, together with companion records (W3C PROV provenance (World Wide Web Consortium, 2013), the DataCite DOI record, and the catalogue submission profile, Table 4), to carry the constructs that have no native QuakeML element: chiefly the composite quality score, cross-catalogue merge lineage, qualitative confidence classes, and macroseismic intensity for historical events. A complete, versioned crosswalk will be published with the schema (see Data and Code Availability) and finalised in Phase 1.

Table 10: Crosswalk from framework catalogue constructs to QuakeML 1.2 / FDSN. Status: **native** = direct QuakeML element; **ext.** = QuakeML **comment** or foreign-namespace extension; **comp.** = carried in a companion record (W3C PROV, DataCite DOI, or the submission profile).

Framework struct	con-	QuakeML 1.2 / FDSN representation	Status
Event and solution identifiers		event@publicID, origin@publicID, magnitude@publicID	native
Origin time and hypocentre		origin/time, /latitude, /longitude, /depth (each value with uncertainty)	native
Location uncertainty / confidence ellipse		origin/originUncertainty (horizontalUncertainty, minHorizontalUncertainty, maxHorizontalUncertainty, azimuthMaxHorizontalUncertainty, confidenceEllipsoid, preferredDescription)	native
Network geometry and fit		origin/quality (azimuthalGap, usedStationCount, usedPhaseCount, standardError)	native
Depth type; method; earth model		origin/depthType, origin/methodID, origin/earthModelID	native
Evaluation mode and status		origin/evaluationMode, origin/evaluationStatus	native
Magnitude (value, type, error, stations)		magnitude/mag (value, uncertainty), magnitude/type, magnitude/stationCount	native
Phase picks and arrivals		pick (time, phaseHint, onset, polarity); arrival (phase, azimuth, distance, timeResidual)	native
Event type and classification		event/type, event/typeCertainty, description	native
Macroseismic intensity (historical)		No native element in QuakeML 1.2 (macroseismic is a QuakeML 2.0 package, still under development); carried via foreign-namespace extension / comment ; intensity recorded as MMI or EMS-98	ext.
Creation provenance		creationInfo (agencyID, author, version, creationTime)	native
Composite quality score (0–100) and grade		No native element; framework-namespace extension on origin , derived from the origin/quality inputs above	ext.
Qualitative confidence class; uncertainty-absent flag		Controlled comment / framework extension; uncertainty_available = false	ext.
Merge lineage / source-event mapping (“derivedFrom”)		No native construct; W3C PROV-O records (World Wide Web Consortium, 2013) and the event-identifier mapping service, referenced from comment	comp.
Catalogue-level descriptors (submission profile)		Minimal native support (eventParameters/description, comment); carried in the submission profile (Table 4)	comp.
Persistent identifier and dataset version		DataCite DOI at dataset level; creationInfo/version per record	comp.

B.4 Versioning Scheme

Table 11 illustrates the version-increment triggers for the three-level MAJOR.MINOR.PATCH scheme described in Section 4.

Table 11: Versioning Scheme: Examples of Version Increment Triggers

Version	Trigger	Example change
MAJOR	Event parameters changed	Reprocessing with new 3D velocity model; recalibration of magnitude scale; new location algorithm affecting systematic offsets
MINOR	New events or fields added	Incorporation of 500 additional aftershock events; addition of focal mechanism fields to existing events
PATCH	Metadata or documentation	Contact email updated; description clarified; deprecated field removed from catalogue-level record

B.5 API Design

A unified API follows REST conventions. Authentication uses token-based credentials for write operations; read operations on public catalogues are unauthenticated. Rate limiting protects service availability, and standard HTTP error codes with a consistent JSON error envelope enable reliable programmatic error handling. Output format is negotiated via the **Accept** header or a **format** query parameter, defaulting to GeoJSON for spatial queries and CSV for tabular exports; QuakeML output is generated from the canonical internal event model, preserving uncertainty fields, evaluation status, and magnitude type. Table 12 summarises the principal endpoint categories.

Table 12: Principal API Endpoint Categories

Category	Operations	Notes
Catalogues	List, retrieve, create, update, delete	Metadata including bounds, event count, licence
Events	List, spatial query, time-range, magnitude/depth filter	Pagination; supports GeoJSON, CSV, QuakeML output
Search	Cross-catalogue full-text and parameter search	Returns ranked results across all accessible catalogues
Import	Upload file, sync from FDSN source, view history	Supports CSV, JSON, GeoJSON, QuakeML 1.2
Merge	Preview, execute, retrieve conflict log	Preview mode returns statistics without writing a new version
Export	Per-catalogue or per-query download	CSV, GeoJSON, KML, QuakeML; version-specific
Versions	List versions, retrieve specific version	Version-specific event snapshots remain available indefinitely

B.6 Access Control and Audit

Access to catalogue data and portal operations is governed by a role-based model. Four *technical* roles are defined (distinct from the governance roles of Section 5.1.3; a single individual may hold

both):

- **Administrator:** full access including user management, system configuration, and deletion of catalogue records.
- **Editor:** can create and modify catalogues, import data, and initiate merge operations, but cannot manage users or alter system configuration.
- **Viewer:** can search, view, filter, and export catalogues to which they have been granted access, but cannot modify data.
- **Guest** (unauthenticated): can discover and download public catalogues; embargoed or restricted catalogues are not visible.

Role assignments are managed by Administrators and recorded in an audit log; a role-upgrade request workflow allows registered users to request elevated access.

Data-level access control. Each catalogue carries one of three access designations: **public** (all, unauthenticated), **registered** (authenticated Viewers and above), or **restricted** (named individuals or groups). This supports time-limited embargoes (e.g. allowing a research group to work privately until a companion paper is published). Field-level masking may be applied where a catalogue contains sensitive location information, such as event locations near culturally significant sites: coordinates may be spatially generalised or withheld for Guest and Viewer access while full precision is retained for authorised users, consistent with the CARE discussion in Section 4.9.

Audit and compliance. All write operations (catalogue creation, import, merge, deletion) are recorded with the authenticated user identity, timestamp, and operation details; read access to restricted catalogues is similarly logged. Where usage agreements are attached to a catalogue (for example, a requirement to cite a companion publication), the agreement is presented at download time and acceptance is recorded.

B.7 Uncertainty Representation

Origin location and time uncertainties are reported as standard deviations where available: **time_uncertainty** (seconds), **horizontal_uncertainty** (kilometres), and **depth_uncertainty** (kilometres). Where horizontal uncertainty is anisotropic, the semi-major and semi-minor axes of the confidence ellipse, its azimuth, and the stated confidence level (conventionally 68%) are reported, mapping to QuakeML **OriginUncertainty** elements (ETH Zürich Seismology / QuakeML Working Group, 2015). Magnitude uncertainties are reported as the standard deviation on the stated scale when available, and the magnitude type code must always accompany the value. Events without quantified uncertainties carry a boolean **uncertainty_available** field set to false; historical events may substitute a qualitative confidence class. Thus the framework requires uncertainty to be represented explicitly, but does not require every legacy or historical record to have a quantified uncertainty. When events are merged, the averaging strategy reports a conservative uncertainty floored at the most precise contributing solution (Section 4.4); priority and quality strategies retain the selected solution’s uncertainty and record alternatives, and the merge log documents the strategy and resulting uncertainty.

B.8 Quality Checks and Full Grade Scheme

Representative automated checks include:

- events ordered chronologically, with no future-dated events;
- depth within a physically plausible range for the region (for New Zealand, about -5 to 700 km, covering Hikurangi–Kermadec subduction depths and allowing for events above the sea-level datum; a higher default applies to global catalogues);
- magnitude within a plausible range for the declared region;
- uncertainty values non-negative and within parameter-specific sanity bounds (for example, horizontal uncertainty not exceeding the catalogue’s declared spatial extent, and magnitude uncertainty below a configurable ceiling such as 1.0 unit);
- no coordinate pair repeated for distinct events at an identical time (duplicate detection);
- station count ≥ 5 for an event to be considered well-constrained; and
- azimuthal gap $< 180^\circ$ for a reliably constrained epicentral location (USGS, 2025).

Results are summarised as passed, warning, or failed. These checks are sanity screens and documentation indicators, not accuracy tests. The full seven-band grade scheme underlying the summary in Table 6 is given in Table 13.

Table 13: Full quality grade thresholds and screening guidance (default scheme; basis and caveat as for Table 6 (Woessner and Wiemer, 2005; Bondár et al., 2004; Warren-Smith et al., 2025)).

Grade	Score	Interpretation and screening guidance
A+	95–100	Very strong documented indicators; still verify suitability for the intended analysis.
A	85–94	Strong documented indicators; suitable for screening and candidate selection.
B+	75–84	Moderate indicators; document limitations in methods sections.
B	65–74	Fair indicators; consider supplementing with additional metadata.
C	45–64	Weak indicators; expert review recommended before quantitative use.
D	35–44	Minimal documented support; use only for discovery or illustrative purposes.
F	< 35	Not recommended for quantitative analysis without substantial review.

B.9 Special Catalogue Type Handling

Regional research catalogues produced with specialised methods (e.g. 3D velocity models, relative relocation, custom magnitude calibrations) are incorporated as distinct catalogues with their own version records and DOIs; processing methods are documented in catalogue-level metadata, and the portal provides cross-reference links between co-temporal events in regional and national catalogues with the spatial and magnitude offset reported. **Temporary deployment catalogues** are incorporated with an explicit time window and a “research” or “monitoring” designation, with station-geometry metadata and links to relevant FDSN waveform services. **Historical and pre-instrumental catalogues** are accommodated through qualitative location confidence classes, macroseismic intensity fields (Modified Mercalli or EMS-98, Grünthal, 1998), narrative source-attribution fields, and separate quality thresholds, and are presented as a separate collection. **Derived and synthetic catalogues** carry mandatory provenance

identifying source catalogue(s), derivation method, parameters, and software, with explicit “derivedFrom” relations to source versions; synthetic catalogues are labelled as such.

B.10 FAIR Compliance Matrix

Table 14 gives the full mapping summarised in Section 4: each FAIR sub-principle, the framework mechanism that addresses it, and a measurable indicator for annual reporting.

Table 14: FAIR compliance matrix: sub-principle, framework mechanism, and indicator.

Sub-pr.	Framework mechanism	Indicator
F1	Persistent identifiers (DOIs) for catalogues; stable event identifiers	% of catalogues with a resolvable DOI; % of events with a stable ID
F2	Rich metadata profile (catalogue- and event-level)	Mean metadata-completeness score across catalogues
F3	Metadata records embed the data identifier	% of metadata records linking to their data identifier
F4	Indexing in the searchable portal and discovery API	% of catalogues indexed and discoverable
A1	Retrieval via FDSN/REST over HTTPS	API uptime; % of catalogues retrievable by identifier
A1.1–2	Open, standard protocols; token auth only for write/restricted data	% of read endpoints requiring no authentication
A2	Metadata and version records retained after data withdrawal	% of withdrawn datasets with surviving metadata
I1	QuakeML / FDSN standard representations	% of catalogues available as valid QuakeML
I2	Controlled parameter lexicon (magnitude, event, phase types)	% of fields using controlled-vocabulary terms
I3	Qualified cross-references (event-ID mapping, lineage links)	% of merged events with recorded source lineage
R1.1	Machine-readable licence on every record	% of catalogues with a machine-readable licence
R1.2	W3C PROV-based provenance (entities, activities, agents)	Mean provenance-completeness score
R1.3	Conformance to FDSN/QuakeML community standards	% of catalogues passing standards-compliance checks

C Summary of Workshop Themes

This appendix summarises the workshop discussions, synthesised here as nine cross-cutting themes. The content reflects community perspectives on minimum catalogue attributes, metadata requirements, processing concerns, governance needs, and priorities for a national FAIR-aligned framework for earthquake catalogues in Aotearoa New Zealand.

Table 15: Summary of Key Themes from the Workshop Discussions

Theme	Key Points
Minimum Catalogue Attributes	Phase picks; station metadata; velocity models (1D/3D); depths and magnitudes with uncertainties; completeness estimates; region of interest; unique IDs; focal mechanisms; processing history; detection and processing methods.
Metadata & Documentation Needs	Clear methodology (location algorithm, velocity model, filters); solution provenance; waveform availability; descriptive summaries; parameter definitions; assumptions (point-source vs 2D/3D); embargo details; QC information; required README and DOI; multiple contacts.
Access & Versioning	Raw data accessibility; open formats (CSV, QuakeML, FDSN); programmatic access (API); search by region, time, magnitude; support for multiple versions; backward/forward compatible formats; tracking of method changes; long-term storage of legacy releases.
Governance Needs	Defined roles (contributor, reviewer, maintainer); central catalogue portal; consistent naming conventions; periodic updates; upload validation; framework transparency; responsibility for long-term stewardship.
Community Priorities	National catalogue portal; common parameter lexicon; national 3D community velocity model; honest uncertainties; digitisation & reprocessing of legacy data; cross-catalogue interoperability; clear documentation for older datasets.
Concerns Raised	Inconsistent completeness; variable processing; missing metadata; uncertain magnitude/depth scales; “black-box” ML outputs; compatibility issues across regional/temporary catalogues; handling of unpublished datasets; need for quality flags and user warnings.
Desired Portal Features	Merge multiple catalogues; support subcatalogs; synthetic catalogues; geographic reference tools; method-based filtering; visualisation; minimum-requirements upload form; linkage among event, waveform, and station data.
What Users Liked	Simple form; unified access; clear uncertainty fields; reproducibility focus; increased awareness of catalogue availability; prioritisation of essential fields.
Requested Improvements	More metadata fields and filters; better uncertainty representation; link with 2D/3D models; clarify naming/versioning; transparent ML integration; improved handling of old locations; inclusion of focal mechanisms and strong-motion connections.

D Glossary

Aggregate assembling many catalogues or agency bulletins into one collection (the ISC/EMSC sense), without necessarily reducing the solutions held for each event.

ANSS / ComCat Advanced National Seismic System; its Comprehensive Earthquake Catalog (USGS). The National Earthquake Information Center (NEIC) is the USGS national hub within ANSS.

API application programming interface: a machine-readable interface for programmatic data access.

Azimuthal gap the largest angle, seen from an epicentre, between adjacent recording stations; a smaller gap means a better-constrained location.

CARE Collective benefit, Authority to control, Responsibility, Ethics: principles for Indigenous data governance.

917 **CMT / GCMT** centroid moment tensor; GCMT is the Global CMT project’s moment-tensor
918 catalogue.

919 **CSV / JSON** comma-separated values; JavaScript Object Notation: plain-text data-interchange
920 formats.

921 **Declustering** separating mainshock–aftershock sequences from background seismicity (e.g. the
922 Gardner–Knopoff method).

923 **DOI** Digital Object Identifier: a persistent, citable identifier for a dataset or document.

924 **ECOS / ECOS-09** Earthquake Catalogue of Switzerland: the SED’s homogenised historical
925 and instrumental catalogue; ECOS-09 is the 2009 release.

926 **EMS-98 / MMI** European Macroseismic Scale (1998); Modified Mercalli Intensity: macroseis-
927 mic intensity scales.

928 **EMSC / ISC** European-Mediterranean Seismological Centre; International Seismological Cen-
929 tre. ISC-GEM is the ISC–Global Earthquake Model (GEM) instrumental reference cata-
930 logue.

931 **EPOS** European Plate Observing System: a coordinated FAIR data infrastructure for solid-
932 Earth science.

933 **FAIR** Findable, Accessible, Interoperable, Reusable: principles for scientific data management.

934 **FDSN** International Federation of Digital Seismograph Networks; its web services are a global
935 standard for seismic data access.

936 **FTE** full-time equivalent (of staff effort).

937 **GeoJSON / KML / GML** open geospatial data formats (JSON-based GeoJSON; Keyhole
938 Markup Language; Geography Markup Language).

939 **GeoNet** New Zealand’s geological-hazard monitoring programme, operated by Earth Sciences
940 New Zealand.

941 **ISF** ISC Standard Format: a seismic-bulletin exchange format.

942 **GSE** Group of Scientific Experts format: a seismic-bulletin exchange format accepted by the
943 ISC.

944 **LFE** low-frequency earthquake: a small, emergent event associated with deep slow slip and
945 tectonic tremor.

946 **Lineage** the explicit record linking a merged or derived event to the specific source events and
947 catalogue versions it was built from; narrower than provenance, which is the full processing
948 history.

949 M_c magnitude of completeness: the lowest magnitude above which all earthquakes in a catalogue
950 are reliably recorded.

951 **Merge** reconciling the multiple solutions different catalogues report for the same earthquake
952 into one unified event (one preferred solution, or all origins retained under one event).

953 M_w , M_L moment magnitude and local (Richter) magnitude: two magnitude scales.

954 **NSHM** (Aotearoa New Zealand) National Seismic Hazard Model.

955 **OBS** ocean-bottom seismometer: a seafloor instrument used in temporary offshore deployments.

956 **PROV / PROV-O** the W3C provenance data model and ontology (entities, activities, and
957 agents).

958 **Provenance** the documented origin and processing history of a dataset (inputs, methods,
959 software, agents).

PSHA probabilistic seismic hazard analysis.

QC quality control.

QuakeML / StationXML standard XML formats for exchanging earthquake parameters and station metadata.

REST representational state transfer: a common web-service architectural style.

RMS root-mean-square (here, of arrival-time residuals): a measure of solution misfit.

SED Swiss Seismological Service.

TVZ Taupō Volcanic Zone.

USGS United States Geological Survey.

UTC Coordinated Universal Time.

W3C World Wide Web Consortium.

WFS Web Feature Service: an Open Geospatial Consortium (OGC) standard for serving geographic features.

WGS84 World Geodetic System 1984: the standard geographic coordinate reference.

References

- Aki, K. (1965). Maximum likelihood estimate of b in the formula $\log n = a - bm$ and its confidence limits. *Bulletin of the Earthquake Research Institute, University of Tokyo*, 43:237–239.
- Aziz Zanjani, F., Lin, G., and Thurber, C. H. (2021). Nested regional-global seismic tomography and precise earthquake relocation along the Hikurangi subduction zone, New Zealand. *Geophysical Journal International*, 227(3):1567–1590.
- Bondár, I., Myers, S. C., Engdahl, E. R., and Bergman, E. A. (2004). Epicentre accuracy based on seismic network criteria. *Geophysical Journal International*, 156(3):483–496.
- Carroll, S. R., Garba, I., Figueroa-Rodríguez, O. L., Holbrook, J., Lovett, R., Materechera, S., Parsons, M., Raseroka, K., Rodriguez-Lonebear, D., Rowe, R., Sara, R., Walker, J. D., Anderson, J., and Hudson, M. (2020). The CARE principles for indigenous data governance. *Data Science Journal*, 19(1):43.
- Chamberlain, C. J., Hopkins, J., Townend, J., van Wijk, K., and Lindsey, N. (2025). Sensing unrest in New Zealand’s largest city: detailed mapping of seismicity in Auckland. Technical Report Report 114, Natural Hazards Commission Toka Tū Ake.
- Christophersen, A., Bourguignon, S., Rhoades, D. A., Allen, T. I., Ristau, J., Salichon, J., Rollins, J. C., Townend, J., and Gerstenberger, M. C. (2024). Standardizing earthquake magnitudes for the 2022 revision of the Aotearoa New Zealand national seismic hazard model. *Bulletin of the Seismological Society of America*, 114(1):111–136.
- DataCite (2025). Versioning. <https://support.datacite.org/docs/versioning>. Guidance on DOI assignment and relations between dataset versions. Accessed: 2026-03-11.
- Diehl, T., Cauzzi, C., Clinton, J., Kraft, T., Kästli, P., Massin, F., Lanza, F., Simon, V., Grigoli, F., Hobiger, M., Roth, P., Haslinger, F., Fäh, D., and Wiemer, S. (2025). Earthquakes in Switzerland and surrounding regions during 2019 and 2020. *Swiss Journal of Geosciences*, 118(15).

Downes, G. L. (1995). Atlas of isoseismal maps of New Zealand earthquakes. Technical report, Institute of Geological & Nuclear Sciences Monograph 11 (EQC Paper 7 / EQC 91/40), Lower Hutt, New Zealand.

Dowrick, D. J. and Rhoades, D. A. (1998). Magnitudes of New Zealand earthquakes, 1901–1993. *Bulletin of the New Zealand Society for Earthquake Engineering*, 31(4):260–280.

Eberhart-Phillips, D. and Reyners, M. (2023). Catalogue of 2001–2011 New Zealand earthquakes relocated with a 3-D seismic velocity model and comparison to 2019–2020 auto-detected earthquakes in the sparsely instrumented southern South Island. *New Zealand Journal of Geology and Geophysics*, 66(4):646–653.

Eiby, G. A. (1968). An annotated list of New Zealand earthquakes, 1460–1965. *New Zealand Journal of Geology and Geophysics*, 11(3):630–647.

Ekström, G., Nettles, M., and Dziewonski, A. M. (2012). The global CMT project 2004–2010: Centroid-moment tensors for 13,017 earthquakes. *Physics of the Earth and Planetary Interiors*, 200–201:1–9.

EPOS (2025a). Epos seismology: Data & services. <https://www.epos-eu.org/tcs/seismology/data-services>. Overview of consolidated seismological data/product services within EPOS ICS. Accessed: 2025-09-30.

EPOS (2025b). Seismology | epos - the european plate observing system. <https://www.epos-eu.org/tcs/seismology>. Describes standardized, FAIR data services provided by the EPOS Seismology Thematic Core Service. Accessed: 2025-09-30.

ETH Zürich Seismology / QuakeML Working Group (2015). Quakeml: A flexible xml representation of seismological data. <https://quake.ethz.ch/quakeml>. QuakeML overview and specification resources. Accessed: 2025-09-30.

European-Mediterranean Seismological Centre (2025a). EMSC homepage. <https://www.emsc.eu/>. Accessed: 2025-10-21.

European-Mediterranean Seismological Centre (2025b). SeismicPortal - web services. <https://www.seismicportal.eu/webservices.html>. Accessed: 2025-10-21.

Fäh, D., Giardini, D., Kästli, P., Deichmann, N., Gisler, M., Schwarz-Zanetti, G., Alvarez-Rubio, S., Sellami, S., Edwards, B., Allmann, B., Bethmann, F., Wössner, J., Gassner-Stamm, G., Fritsche, S., and Eberhard, D. (2011). ECOS-09 earthquake catalogue of Switzerland: Release 2011 report and online catalogue. Technical Report SED/RISK/R/001/20110417, Swiss Seismological Service, ETH Zürich. Accessed: 2026-03-12.

Fasola, S. L., Jackson, N. M., and Williams, C. A. (2023). Deep Short-Term Slow Slip and Tremor in the Manawatu Region, New Zealand. *Geophysical Research Letters*, 50(17).

Gardner, J. K. and Knopoff, L. (1974). Is the sequence of earthquakes in southern California, with aftershocks removed, Poissonian? *Bulletin of the Seismological Society of America*, 64(5):1363–1367.

GeoNet (2025a). Earthquake catalogue. https://www.geonet.org.nz/data/types/eq_catalogue. Describes the GeoNet catalogue, Public ID policy, and access tools (Quake Search, WFS, FDSN, API). Accessed: 2025-09-30.

GeoNet (2025b). Earthquake location information (seiscomp, locsat, nonlinloc, magnitudes). https://www.geonet.org.nz/data/supplementary/earthquake_location. Explains location algorithms, velocity models, and summary magnitude computation at GeoNet. Accessed: 2025-09-30.

- GeoNet (2025c). Fdsn webservice (event, station, waveform). <https://www.geonet.org.nz/data/access/FDSN>. Details GeoNet’s FDSN services and QuakeML output; notes eventtype parameter. Accessed: 2025-09-30.
- GeoNet (2025d). Geonet api. <https://api.geonet.org.nz/>. Landing page for the GeoNet API and documentation. Accessed: 2025-09-30.
- Geoscience Society of New Zealand (2023). Pre-conference workshops (gsnz 2023): Framework development for earthquake catalogues in new zealand. <https://confer.eventsair.com/gsnz2023/workshops>. Workshop page outlining aims to develop a comprehensive framework and improve FAIRness. Accessed: 2025-09-30.
- Geoscience Society of New Zealand (2024). Pre-conference workshops (gsnz 2024): Next steps in building a framework for earthquake catalogues in aotearoa new zealand. <https://confer.eventsair.com/gsnz2024/workshops>. Workshop page summarising progress and efforts to improve FAIRness; includes session description. Accessed: 2025-09-30.
- Gerstenberger, M. C. et al. (2024). The 2022 Aotearoa New Zealand national seismic hazard model: Process, overview, and results. *Bulletin of the Seismological Society of America*, 114(1):7–36.
- GNS Science (2020). National earthquake information database. <https://www.gns.cri.nz/data-and-resources/national-earthquake-information-database/>. Accessed: 2025-10-21.
- GO FAIR Initiative (2025). Fair principles. <https://www.go-fair.org/fair-principles/>. Accessed: 2026-03-11.
- Grünthal, G., editor (1998). *European Macroseismic Scale 1998 (EMS-98)*, volume 15. Cahiers du Centre Européen de Géodynamique et de Séismologie, Luxembourg.
- Illsley-Kemp, F. and Mestel, E. (2025). A new consistent and high-precision earthquake catalogue for the Taupō Volcanic Zone, New Zealand. *Seismica*, 4(1).
- International Seismological Centre (2025a). Data collection. <https://www.isc.ac.uk/standards/datacollection/>. Accessed: 2025-10-21.
- International Seismological Centre (2025b). The reviewed ISC bulletin. <https://www.isc.ac.uk/iscbulletin/review/>. States the reviewed bulletin is typically 24 months behind real-time. Accessed: 2026-06-18.
- International Seismological Centre (2025c). Standard bulletin search. <https://www.isc.ac.uk/iscbulletin/search/bulletin/>. Accessed: 2025-10-21.
- Kukutai, T. and Taylor, J., editors (2016). *Indigenous Data Sovereignty: Toward an Agenda*. CAEPR Research Monograph No. 38. ANU Press, Canberra.
- Michailos, K., Chamberlain, C. J., Simpson, G., Cox, S. C., Townend, J., Vargo, L. J., Oestreicher, N., and Miller, M. S. (2025). Temporal Evolution of Seismicity in the Central Southern Alps, New Zealand: Evidence for Rainfall-Triggered Seismicity. *Geochemistry, Geophysics, Geosystems*, 26.
- Michailos, K., Smith, E. G. C., Chamberlain, C. J., Savage, M. K., and Townend, J. (2019). Variations in Seismogenic Thickness Along the Central Alpine Fault, New Zealand, Revealed by a Decade’s Relocated Microseismicity. *Geochemistry, Geophysics, Geosystems*, 20(1):470–486.

1084 Mulder, T. (2016). Coordination of event type codes and their relation to quakeml. https://brtt.com/wp-content/uploads/2023/04/events/alaska2016/talks/Mulder_Event_types.pdf.
1085 Slides discussing QuakeML 1.2 event type classifications and related coding practices. Accessed:
1086 2025-09-30.
1087

1088 New Zealand Government (2022). New zealand government open access and licensing framework
1089 (NZGOAL). <https://data.govt.nz/manage-data/policies/nzgoal>. Accessed: 2026-03-11.

1090 Northern California Earthquake Data Center (2024). ANSS composite catalog search help.
1091 <https://www.ncedc.org/anss/catalog-search-help.html>. Documents the 16-second /
1092 100-km duplicate-detection thresholds used in the ANSS Composite Catalog. Accessed: 2026-
1093 03-29.

1094 Rollins, C., Christophersen, A., Thingbaijam, K. K. S., Gerstenberger, M. C., Hutchinson, J.,
1095 Eberhart-Phillips, D., Bannister, S., Van Dissen, R. J., Seebeck, H., and Ellis, S. (2025). An
1096 integrated earthquake catalog for Aotearoa New Zealand. *Bulletin of the Seismological Society*
1097 *of America*, 115(4):1703–1718.

1098 Si, Z., Zhuang, J., Gentili, S., Jiang, C., and Wang, W. (2024). A Bayesian merging of
1099 earthquake magnitudes determined by multiple seismic networks. *Seismological Research*
1100 *Letters*, 95(6):3221–3233.

1101 U.S. Geological Survey (2025). Fdsn event web service specification — usgs implementation docs.
1102 <https://earthquake.usgs.gov/fdsnws/event/1/>. Reference for FDSN Event web service
1103 parameters and usage. Accessed: 2025-09-30.

1104 USGS (2025). ANSS comprehensive earthquake catalog (ComCat) documentation. <https://earthquake.usgs.gov/data/comcat/>. Accessed: 2025-10-21.
1105

1106 Vorobieva, I. A., Gvishiani, A. D., Dzeboev, B. A., Dzeranov, B. V., Barykina, Y. V., and
1107 Antipova, A. O. (2022). Nearest neighbor method for discriminating aftershocks and duplicates
1108 when merging earthquake catalogs. *Frontiers in Earth Science*, 10:820277.

1109 Wallace, L. M. and Beavan, J. (2010). Diverse slow slip behavior at the Hikurangi subduction
1110 margin, New Zealand. *Journal of Geophysical Research: Solid Earth*, 115(B12).

1111 Warren-Smith, E., Jacobs, K., Rollins, C., Chamberlain, C. J., Eberhart-Phillips, D., and
1112 Williams, C. (2025). A quantitative assessment of GeoNet earthquake location quality in
1113 Aotearoa New Zealand. *New Zealand Journal of Geology and Geophysics*, 68(5):941–954.

1114 Wiemer, S. and Wyss, M. (2000). Minimum magnitude of completeness in earthquake catalogs:
1115 Examples from Alaska, the western United States, and Japan. *Bulletin of the Seismological*
1116 *Society of America*, 90(4):859–869.

1117 Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A.,
1118 Blomberg, N., Boiten, J.-W., da Silva Santos, L. B., Bourne, P. E., Bouwman, J., Brookes,
1119 A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R.,
1120 Gonzalez-Beltran, A., Gray, A. J. G., Groth, P., Goble, C., Grethe, J. S., Heringa, J., 't Hoen,
1121 P. A. C., Hooft, R., Kuhn, T., Kok, R., Kok, J., Lusher, S. J., Martone, M. E., Mons, A.,
1122 Packer, A. L., Persson, B., Rocca-Serra, P., Roos, M., van Schaik, R., Sansone, S.-A., Schultes,
1123 E., Sengstag, T., Slater, T., Strawn, G., Swertz, M. A., Thompson, M., van der Lei, J., van
1124 Mulligen, E., Velterop, J., Waagmeester, A., Wittenburg, P., Wolstencroft, K., Zhao, J., and
1125 Mons, B. (2016). The FAIR guiding principles for scientific data management and stewardship.
1126 *Scientific Data*, 3:160018.

- 1127 Woessner, J., Hardebeck, J. L., and Hauksson, E. (2010). What is an instrumental seismicity
1128 catalog. *Community Online Resource for Statistical Seismicity Analysis*. Accessed: 2025-10-21.
- 1129 Woessner, J. and Wiemer, S. (2005). Assessing the quality of earthquake catalogues: Estimating
1130 the magnitude of completeness and its uncertainty. *Bulletin of the Seismological Society of*
1131 *America*, 95(2):684–698.
- 1132 World Wide Web Consortium (2013). Prov-o: The prov ontology. [https://www.w3.org/TR/
1133 prov-o/](https://www.w3.org/TR/prov-o/). W3C Recommendation. Accessed: 2026-03-11.
- 1134 Yarce, J., Sheehan, A. F., Nakai, J. S., Schwartz, S. Y., Mochizuki, K., Savage, M. K., et al.
1135 (2019). Seismicity at the Northern Hikurangi Margin, New Zealand, and Investigation of the
1136 Potential Spatial and Temporal Relationships With a Shallow Slow Slip Event. *Journal of*
1137 *Geophysical Research: Solid Earth*, 124(5):4751–4766.